

Staffing Levels for European Healthcare

Report by the Expert Group on
Health Systems Performance Assessment



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TABLE OF CONTENTS

LIST OF FIGURES, BOXES, TABLES AND ANNEXES.....	3
ACKNOWLEDGEMENTS.....	5
EXECUTIVE SUMMARY	6
KEY MESSAGES.....	8
Introduction.....	10
Background of the study	10
Aim and structure.....	12
Methodology.....	12
Chapter 1: Overview of results.....	14
Chapter 2: What are Member States' objectives and goals when defining staffing levels?.....	16
System level.....	16
Sectoral level.....	18
Hospitals.....	18
Primary care.....	19
Other sectors.....	20
Chapter 3: What kinds of defined staffing level solutions are in place?.....	21
System level.....	21
Sectoral level.....	27
Approach to staffing levels in hospitals.....	27
Approach to staffing levels in primary care.....	30
Approach to staffing levels in other sectors (ambulatory, specialist and dental care).....	32
Chapter 4: How are staffing levels governed?.....	33
System level.....	33
What are the governance solutions, how are the staffing levels agreed upon and who is involved in the agreement process?.....	33
Is there a link between staffing levels and HWF planning, investment and training capacities?.....	36
Are staffing levels legally binding?.....	42
Sectoral level: hospitals, primary care and other sectors.....	46
Is there a link between hospital staffing levels and HWF planning, investment and training capacities?.....	49
Are staffing level solutions in primary care an emerging and promising trend?.....	50
Chapter 5: What do we know about the evaluation of staffing levels?.....	51
System level.....	51
Do countries see opportunities and risks of setting staffing levels at the system level?.....	51
Sectoral level.....	60
Do countries see opportunities and risks of setting staffing levels at the sectoral level?.....	60
Do countries evaluate their staffing levels at the sectoral level?.....	63
Conclusions: What considerations do you have to think about when implementing or changing staffing levels?.....	67
REFERENCES.....	69
ANNEXES.....	70

LIST OF FIGURES, BOXES, TABLES AND ANNEXES

Figures

- Figure 1 Questionnaire timeline and response rate
- Figure 2 EU Member States and Norway reporting for staffing levels
- Figure 3 Staffing levels at the sectoral level by country
- Figure 4 Reasons for defining staffing levels at the system level
- Figure 5 Reasons for defining staffing levels at the sectoral level for hospitals
- Figure 6 Reasons for defining staffing levels at the sectoral level for primary care
- Figure 7 Staffing level solutions in place at the system level by country

Boxes

- Box 1 Definitions of different approaches to defining staffing levels
- Box 2 Definitions of declared objectives and goals for defining staffing levels
- Box 3 Czechia uses fixed staffing levels to achieve clear, quantifiable targets for health professionals
- Box 4 Ireland uses flexible staffing to align the health workforce with system needs
- Box 5 Malta's evidence-based workforce planning approach
- Box 6 Norway opts for quality over quotas
- Box 7 Flexibility over ratios: Finland's alternative to staffing levels
- Box 8 Sweden's alternative to staffing levels
- Box 9 Austria uses layered legal and planning tools to define hospital staffing levels
- Box 10 Balancing fixed rules with budget-linked flexibility in Luxembourgish hospital staffing
- Box 11 Flexibility and decentralisation drive Spain's hospital workforce strategy
- Box 12 Portugal balances fixed nursing ratios with flexible team-based allocation
- Box 13 How Italy governs health staffing
- Box 14 Stakeholders agreements on staffing levels in Bulgaria
- Box 15 Romania's health workforce approach connects planning and staffing
- Box 16 How staffing levels are linked with investment decisions in Poland
- Box 17 From policy to practice: How Latvia links staffing levels and education
- Box 18 How Greece links Workforce Planning and Staffing Levels
- Box 19 Linking staffing levels with training, investment, and workforce planning in Spain
- Box 20 How Hungary embeds staffing levels in law and policy
- Box 21 The strategic influence of non-binding staffing solutions in Portugal
- Box 22 Governance of staffing levels in Belgian hospitals
- Box 23 Germany's governance model for hospital staffing levels
- Box 24 The governance of staffing levels in Slovakian primary care
- Box 25 Estonia's path toward possible staffing levels
- Box 26 Workload data as the basis for future staffing in Slovenia
- Box 27 A legally anchored system-level evaluation approach in Poland
- Box 28 Institutionalised evaluation through the Romanian National Health Institutes

Box 29 Evaluating Safe Nurse Staffing in Ireland

Box 30 Evaluating Minimum Nurse Staffing Regulations in Germany

Box 31 Expert-Based and emerging approaches to staffing level evaluations in Bulgaria

Box 32 Ireland's research-based framework for safe nurse staffing and skill mix

Box 33 Austria's structural plan for primary care staffing is a balance between flexibility and consistency

Box 34 Evaluating hospital staffing levels in Greece

Box 35 From fixed ratios to evidence-based evaluation in Germany

Box 36 Hungary's centralised evaluation of primary care staffing

Tables

Table 1 Overview of countries that filed out the questionnaires

Table 2 Opportunities and future staffing level plans

Annexes

Annex 1 – Main conclusions from the Policy Focus Group

Annex 2 – Survey questionnaires

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This publication is a report by the European Commission's Member States Expert Group on Health Systems Performance Assessment (HSPA EG). Observers in this group include Norway, the Organisation for Economic Co-operation and Development (OECD) Health Division, the World Health Organization (WHO) Regional Office for Europe, and the European Observatory on Health Systems and Policies. The report is based on an analysis of replies to a survey developed and conducted within the Expert Group on HSPA. The content of the report greatly benefited from the ideas, feedback, reflections, experiences and suggestions provided by members of the HSPA EG and particularly by the members of the HSPA working group on healthcare workforce, as well as by the European Commission (DG SANTE) healthcare workforce team.

This report was written by Michelle Falkenbach, Gemma Williams and Matthias Wismar (European Observatory on Health Systems and Policies), with the support of the HSPA Secretariat in DG SANTE, led by Barbara Lipszyc.

The country case study boxes were prepared by the authors on the basis of the survey results. Members of the HSPA Expert Group and the HSPA working group on healthcare workforce were given the opportunity to review, validate and refine the case descriptions. Any remaining inaccuracies or misinterpretations are the responsibility of the authors.

The report was developed under the supervision of the co-chairs of the Expert Group on HSPA, Dirk Van den Steen (European Commission, DG SANTE) and Kenneth Grech (Malta), and of the rapporteur of the HSPA working group on health workforce, Tomáš Troch (Czechia).

EXECUTIVE SUMMARY

In 2024, the Expert Group on Health Systems Performance Assessment (HSPA EG) identified health workforce as a strategic priority, highlighting that appropriate staffing levels are fundamental to resilient, safe, and high-performing health systems. The choice of this topic builds on discussions with the HSPA EG around existing EU health workforce actions and is intended to complement them.

The HSPA EG, supported by the European Observatory on Health Systems and Policies, undertook an in-depth analysis of how EU Member States and Norway define, implement, and use staffing levels. This report draws on three main sources of evidence: Member State responses to two surveys (system and sectoral levels), the October 2025 workshop on *Staffing Levels for Healthcare*, and a Policy Focus Group held in November 2025.

Health systems perform best when the right number and mix of health professionals are available in the right places and at the right time. Adequate staffing is directly linked to quality, safety, access, and staff well-being. While the evidence base supporting these links is strong, countries differ markedly in how staffing levels are defined, implemented, and monitored, reflecting variation in population needs, healthcare organisation, workforce composition, and available technology.

Countries adopt diverse approaches, ranging from fixed ratios to flexible models and hybrid systems. Fixed ratios offer predictability and transparency, often anchored in law or collective agreements, while flexible models allow responsiveness to demographic change, local demand, and technological development. Many systems now seek to balance these through adaptive frameworks that combine minimum thresholds with data-driven planning and workload-based methodologies. Across the EU, there is growing momentum toward linking staffing levels with forecasting and planning tools that anticipate future workforce needs and service demand.

The survey revealed that staffing policies are primarily driven by health system goals, improving quality, safety, and workforce protection, rather than broader societal objectives such as economic growth or environmental sustainability. However, ensuring safe staffing also supports wider social and economic stability, especially in underserved or remote regions.

At the sectoral level, hospitals remain the main area where staffing ratios are applied, reflecting their structured environments and measurable workloads. Safety and effectiveness are the dominant goals. In primary care, fewer countries define ratios, instead adopting flexible, multidisciplinary, and regionally tailored approaches to address shortages and support integrated community care. Ambulatory and specialist care rely on adaptable, demand-based models that can respond to local needs and service diversity.

Governance of staffing levels is typically shared among multiple actors: ministries of health lead, but ministries of finance and education, professional bodies, health insurers, and social partners all play crucial roles. This multilevel collaboration ensures alignment between staffing, training, and financing. Increasingly, countries employ analytical tools such as population and workload profiling, forecasting, and skill-mix modelling to inform decisions and evaluate their effects. Where staffing levels are legally binding, they are often monitored through licensing, payer contracts, or inspections. Where they are non-binding, they still function as reference standards guiding planning, budgeting, and workforce allocation.

Links to health workforce planning, investment, and training are increasingly evident. Many Member States use staffing rules as a lever to adjust medical and nursing education capacity, steer financial allocation, and anticipate future shortages. This alignment between staffing and planning enhances policy coherence and system resilience.

Overall, the report identifies a clear shift away from rigid rules toward adaptive, evidence-based staffing frameworks. These approaches combine minimum protections with flexibility through task-shifting, evolving competencies, regional tailoring, and continuous evaluation. While EU-wide staffing benchmarks are neither feasible nor desirable given system diversity, exchange of practices and cooperation on methodologies, indicators, and evaluation tools can strengthen comparability, learning, and accountability.

Staffing levels are not merely an operational parameter but a strategic policy lever linking workforce availability with health system performance. Using them effectively requires balancing assurance with adaptability, grounded in robust data, transparent governance, and a shared commitment to quality, safety, and workforce well-being across Europe's health systems.

KEY MESSAGES

- **Health system priorities drive staffing policies.**
Countries define staffing levels primarily to improve patient safety, access to care, and workforce well-being. Broader societal goals such as economic development or sustainability tend to play a secondary role. A supported and adequately staffed health workforce remains fundamental to achieving strong population health outcomes.
- **Technological, demographic, and epidemiological differences shape staffing needs.**
For many countries, rigid or uniform ratios risk being impractical and incompatible with national and local systems due to variations in population health, service organisation, composition of health workforce and available technology.
- **Staffing standards should be adaptable and responsive to various contexts.**
Staffing levels should be defined by and responsive to each country's health needs, organisational structure, and financial capacity rather than imposed through a one size-fits-all approach at EU level.
- **Fixed staffing ratios are less commonly applied in primary care, which requires flexible, multidisciplinary staffing approaches.**
In the context of workforce shortages, diverse professional roles, and regional variation in capacity, fixed ratios can reinforce existing constraints. While patient-capitated, case-mix-adjusted ratios could theoretically enable expansion, most systems rely instead on flexible, multidisciplinary team models. These models allow roles and workloads to be adjusted according to local workforce availability and population needs, making primary care particularly unsuited to rigid staffing norms.
- **Nursing roles and staffing standards vary across contexts.**
Differences in roles, competencies, and responsibilities mean that staffing levels are defined and applied differently across nursing professions and systems.
- **Governance is shared across multiple actors.**
Although Ministries of Health usually lead on staffing policies, governance is complex. Finance and education ministries, health insurers, professional bodies, and social partners all co-shape staffing decisions, balancing service needs with budgetary constraints, training capacity and workforce expectations.
- **Health workforce planning is an underused lever for staffing policy.**
Many countries already use workforce planning and forecasting to inform staffing levels, but these tools could be scaled up and used more strategically. Better forecasting of healthcare demand can generate more reliable estimates of workforce requirements, support innovative care models and strengthen staffing in community and primary care settings while underscoring the need for adaptable staffing approaches.
- **Adaptive approaches are replacing rigid rules.**
Countries are increasingly combining minimum standards with flexible safeguards to make staffing policies more responsive and sustainable. Examples include task-shifting, evolving competencies, regional tailoring, and periodic review mechanisms that allow systems to adjust over time.

- **Evaluating staffing levels is essential to understand whether they deliver their intended impact.**

Regular assessment is critical to determine whether staffing standards improve quality of care, enhance efficiency, support workforce retention, and strengthen overall health system performance.

- **EU-wide benchmarks are not feasible, but shared learning through comparable evidence is essential.**

Given the diversity of European health systems, uniform minimum staffing benchmarks at EU level are neither practical nor desirable. However, structured exchange of practices and cooperation on common methodologies, data standards, and evaluation tools can deliver substantial benefits. Developing comparable indicators and quality metrics would enable better monitoring of how staffing approaches affect patient safety and system performance, strengthening shared learning and accountability across countries.

Introduction

Background of the study

In 2014, the European Commission established the Expert Group on Health Systems Performance Assessment⁽¹⁾ (HSPA EG) to provide European Union countries with a dedicated forum to exchange experiences and to support national policymakers by identifying tools and methodologies to develop HSPA.

Each Member State can appoint up to two representatives as members of the Expert Group, and the European Commission is a member too. Norway and international organisations such as the World Health Organisation (WHO), the European Observatory on Health Systems and Policies and the Organisation for Economic Co-operation and Development (OECD), participate as observers, actively contributing to the work of the Group.

The HSPA EG defines priority areas that it wishes to work on. Since July 2022, the mandate of the Expert Group on HSPA has been expanded beyond HSPA, providing EU countries with a forum to discuss and promote strategic innovative approaches to strengthening health systems.

Within this expansion, healthcare workforce, with a focus on staffing levels, was selected as the priority topic for 2025. This focus prompted dedicated discussions and presentations on EU level actions addressing health workforce challenges. To support this work, the Expert Group agreed to conduct a survey on existing staffing levels and their impact on policies aimed at addressing health workforce shortages, to organise a dedicated workshop and to produce an HSPA EG report on this topic. The European Observatory on Health Systems and Policies was invited to support this work.

Accordingly, this report is informed by three complementary sources of qualitative evidence: i) the analysis of HSPA EG Members' replies to the two questionnaires (discussed further below)⁽²⁾, ii) the outcome of a workshop on Staffing Levels for Healthcare that was held on 16 October 2025 in Brussels, and iii) the feedback received from a Policy Focus Group. The conclusions from the Policy Focus Group are reported in Annex 1.

High-performing health systems rely on having the right number and mix of health workers in place to achieve health system goals and objectives, such as those related to health outcomes, staff and patient satisfaction, access, and quality. Shortages or imbalances in staffing can lead to delays, unsafe treatments, and poor user experiences, especially in ambulatory care clinics, emergency departments, and primary care settings to name a few. In the frame of Health System Performance Assessments (HSPA), staffing levels can be used as a set of representative measures to link workforce availability to broader health system objectives. International research consistently shows that staffing levels are closely linked to a range of health system outcomes-including patient safety, staff wellbeing, access to care, and overall quality.

Staffing ratio policies have been implemented for longer in countries beyond the EU countries, providing a wider and more mature evidence base to inform their evaluation. A study by Sun et al., (2024) found that higher clinician staffing ratios in US health centres were associated with improved performance across multiple quality indicators, including vaccination rates,

(1) https://health.ec.europa.eu/health-systems-performance-assessment/overview_en#expert-group.

(2) After the questionnaires were submitted some countries requested small edits. The edits requested by Latvia and Poland were a bit bigger.

cancer screenings, and mental health outcomes. Similarly, a systematic review by Dall’Ora et al., 2022 determined that higher nurse staffing levels were linked with improved patient outcomes including lower mortality, fewer complications and shorter length of stay in hospital. Evidence on California’s mandated nurse-to-patient ratios further shows an association between higher ratios and significantly reduced mortality and adverse events, alongside improving nurse satisfaction and retention (Aiken et al., 2010).

Beyond the United States, Australia and New Zealand have established structured, data-driven staffing models that illustrate this approach. Victoria was the first Australian state to legislate nurse-to-patient ratios in 2000, later strengthened through the Safe Patient Care Act 2015 (Government of Victoria, 2015), with similar frameworks introduced in Queensland and New South Wales (NSW Health, 2022). New Zealand’s Care Capacity Demand Management (CCDM) programme provides a national, evidence-based model linking patient acuity, workload measurement, and staffing forecasts that ensures consistency and quality while maintaining flexibility at the local level.

In the United Kingdom, staffing levels are embedded in professional and regulatory standards that support transparency and system accountability. The National Institute for Health and Care Excellence (NICE) provides evidence-based guidance on safe staffing (NICE, 2014).

This report presents the overview of staffing level solutions in the EU countries and Norway. Some of them offer examples of how staffing ratios have been evaluated. Stepping up efforts in health workforce planning, supported by the Joint Action HEROES (HEalth woRkfOrce to meet health challEnges)⁽³⁾, is an opportunity to advance and refine staffing level solutions, ensuring they are evidence-based, better aligned with service needs, and more resilient to future workforce pressures.

As countries embark on Health System Performance Assessments (HSPA) it would be beneficial to identify one or several indicators that help systems understand the link between workforce availability and broader health system objectives and highlight areas that require further investigation. These indicators can serve as a bridge between performance monitoring and health workforce planning, ensuring that staffing levels are not only tracked but also aligned with future service delivery needs. For instance, in Czechia’s HSPA framework, indicators explicitly refer to “future workforce needs”, thereby embedding workforce planning into the performance assessment process. This approach demonstrates how HSPA can provide actionable intelligence: it links data on current staffing levels with projections of demand, creating a structured mechanism to anticipate gaps and guide policy responses. By doing so, countries can strengthen the connection between assessing health system performance, planning for the workforce of tomorrow, and ensuring that staffing levels remain responsive to evolving health priorities.

Currently, many European countries have or are seeking to define and implement staffing standards across various sectors of their health systems. However, the terminologies and approaches used to establish these standards may differ between (and within) countries and there is little systematic knowledge about what countries are doing or about the emerging trends in staffing-related reforms.

(3) The main aim of this joint action is to improve the countries’ capacity for health workforce planning to ensure accessible, sustainable and resilient health systems. See <https://healthworkforce.eu/> for more information.

Aim and structure

This analysis is informed by a two-questionnaire survey, one on system level staffing levels and the other on sectoral (hospital, primary care, ambulatory care, etc.) staffing levels. The aim of the survey is to map approaches and methods to address staffing levels in the healthcare workforce (HWF) in EU Member States and in Norway. By understanding how staffing levels are defined (Box 1), implemented, utilised and evaluated across countries, there is an opportunity to facilitate mutual learning and identify best practices.

The study aims to answer the following questions:

- Do countries use staffing levels in their health systems (either at the system level or at the sectoral level)?
- How are staffing levels governed (from definition to implementation and then monitoring)?
- What do we know about the evaluation of staffing levels?
- How do regulations and guidelines on staffing levels link to HWF planning, investment and training capacities?

The remainder is organised as follows. After the methodology section, the results are presented in five chapters: Overview; The objectives and goals when defining staffing levels; Types of staffing levels in place; Governance; Evaluations and future outlook. Finally, the conclusions address implementation and policy transfer.

BOX 1- Definitions of different approaches to defining staffing levels

Countries adopt different approaches to defining staffing levels:

Fixed ratio staffing levels usually predefined ratios, typically specifying the number of health workers or professionals relative to patients, population groups, hospital beds, geographical areas. These ratios may be legally mandated or outlined in recommendations or guidelines.

Flexible staffing models are defined based on a combination of supply and demand factors aligning health worker availability with the healthcare needs of the population. This approach offers flexibility to adjust to workforce dynamics and evolving needs.

Other countries have no defined staffing levels. In some countries staffing levels are not formally defined, and neither fixed ratios nor flexible staffing models are used.

Minimum threshold is the lowest number of healthcare workers that is deemed necessary to maintain safe, effective, and sustainable service delivery. This can include thresholds in the form of aspirational targets, guiding principles and mandatory measures.

Methodology

Between February and June 2025, the European Observatory, the HSPA Expert Group and the HSPA Secretariat, designed, piloted, and distributed two questionnaires to 27 EU Member States and Norway, 28 in total (Figure 1). The questionnaires were developed following several rounds with the HSPA working group on healthcare workforce. It was emphasised that the questionnaires should map different healthcare sectors (e.g. hospital, primary care, etc.) and distinguish between doctors and nurses, as well as consider other professions.

Both questionnaires examine whether and how countries set staffing levels. One questionnaire looks exclusively at the system level, while the other one looks towards the sectoral level (hospitals, primary care, ambulatory care, etc.). For the sectoral level questionnaire, Member States had the opportunity to fill in several versions of the questionnaire for different sectoral levels. For example, country x will have filled out a system level questionnaire and then a sectoral level questionnaire for hospitals and a sectoral level questionnaire for primary care. All Member States filled out system level questionnaires, but not all Member States filled out sectoral level questionnaires. The number of sectoral questionnaires a Member State completed was left up to the responding Member State to decide.

Figure 1 Questionnaire timeline and response rate



Limitations

One of the limitations of the survey and thus this analysis is that some countries did not fill in any sectoral questionnaires. This could be due to several reasons: a) they did not have time, b) there were no experts available in the time given to fill out the questionnaire, c) there was no interest in filling out the questionnaire, d) they had put sectoral information into the system questionnaire and did not want to duplicate information, d) there was nothing to report on any sectoral level, or e) the sectoral questionnaires might be incomplete in the sense that some Member States might have focused their response on hospitals and in doing so might have skipped filling out a sectoral questionnaire for ambulatory care even though there might have been something relevant to report.

It should also be acknowledged that, beyond differences in staffing levels and professional classifications, the definition, scope and boundaries of health sectors themselves may vary across countries. Structural differences in how countries organise and categorise hospital, ambulatory, specialist or primary care services may affect how responses were interpreted and reported. These variations limit the direct comparability of findings related to distinct health system sectors and should be taken into account when drawing policy-relevant conclusions.

It must also be noted that the HSPA Expert Group agreed on a scope that did not include long term care (LTC). This was done primarily because for many countries LTC area rests within the social ministries and not within the Ministry of Health, and therefore outside the scope of this research.

Chapter 1: Overview of results

The questionnaire was sent to a total of 28 countries including all EU Member States and Norway. The response rate was 78% with 22 countries completing questionnaires. A total of 52 questionnaires were received counting all completed questionnaires from the system and sectoral levels (Table 1).

In total, 9 responding countries indicated that they define staffing levels at the system level and 12 at the sectoral level⁽⁴⁾ (Table 1 & Figure 2). The number of countries that define staffing levels at the sectoral level (for which they filled in a questionnaire) are as follows: hospitals (12), primary care (6), ambulatory care (3), specialist care (1), dental care (0)⁽⁵⁾ (Figure 3). There are countries that filled out a questionnaire, for example for dental care, to clarify that they do not define staffing levels at this sector.

Table 1 Overview of countries that filed out the questionnaires and define staffing levels

Country	System Level	Hospital	Primary Care	Ambulatory Care	Specialist Care	Dental Care
Belgium	no	yes				
Bulgaria	yes	yes	yes			
Czechia	yes	yes				
Germany	no	yes				
Estonia	no					
Ireland	yes	yes				
Greece	yes	yes				
Spain	no	yes	yes		yes	
Italy	no		no			
Latvia	yes					
Luxembourg	no	yes				
Hungary	yes	yes		yes		
Malta	no					
Austria	no	yes	yes	yes		
Poland	yes					
Portugal	yes	yes	yes			
Romania	yes					
Slovenia	no					
Slovakia	no	yes	yes	yes		
Finland	no	no	no	no		
Sweden	no					
Norway	no		no		no	no

Note: The indication of “yes” or “no” pertains to whether that country defines staffing levels at that level. If the box is blank, no questionnaire was received.

(4) Some Member States filled out information for the sectoral level in the system level questionnaire and therefore did not fill out an extra questionnaire for the sectoral level.

(5) One country filled in a questionnaire for long term care and nursing, but it is out of the defined scope.

Reporting

- Reported using staffing levels at any level
- Did not report using staffing levels
- Included in the survey but did not respond
- Data not available

0 250 500 1000 km

Hospital

- Fixed and flexible
- Fixed
- Flexible

Primary care

- Fixed and flexible
- Flexible

Ambulatory care

- Fixed and flexible

Specialist care

- Flexible

0 125 250 500 km

Chapter 2: What are Member States' objectives and goals when defining staffing levels?

System level

The eight countries that stated they defined staffing levels at the system level were asked their reasons for doing so⁽⁶⁾. The options suggested as reasons can be found in Box 2.

BOX 2 Definitions of declared objectives and goals for defining staffing levels

1. Health System–Related Goals

Countries want staffing approaches to help the health system perform better and be more resilient. These goals typically include people-centeredness, health improvement, and financial protection and they are the outcomes used to measure the performance of a health system. They are very relevant to patients and citizens. The health workforce influences these final goals by impacting the intermediate objectives.

Effectiveness of care: The ability of an intervention to have a meaningful effect on patients in normal clinical conditions.

Safety of care: The prevention of errors and adverse effects associated with health services.

User experience: Users' beliefs, preferences, perceptions, responses, and behaviours that occur before, during and after.

Access to care: The opportunity or ease with which individuals or communities are able to use appropriate health services.

Patient centeredness: As a goal which captures how far the health system adequately addresses people's non-medical health needs.

Health improvement: A fundamental goal of any health system. It encompasses the health of the population at different points in the life cycle and is measured through morbidity and mortality indicators as well as aggregate life expectancy.

Financial protection: Refers here to the health system's role in protecting the population against risk. For example, if there are not enough doctors working in the public sector, people may be forced to visit a doctor in the private sector, leading to out-of-pocket spending.

Quality of care: is used as a generic term comprising various dimensions including e.g. access, safety, effectiveness, user experience but also efficiency and equity.

(6) Question 1.2 instructions: In the Box below (Box 2) several goals are defined. These can be the reasons for defining staffing levels in a country/region. We have identified three broad categories:

- Health system objectives and goals
- Health workers related goals
- Societal goals.

To answer the following questions, please refer to official documents such as policies, legislation, and guidelines. These may be issued by the government, the Ministry of Health, competent arm's length bodies, social partners delegated to draft such documents. If, for example, effectiveness of care is an official reason as to why your country/ region is defining staffing levels, please check the box next to it.

Several official goals can be reported. If the documents you have identified include additional reasons for defining staffing levels, please specify these in your response. Please include in writing or provide a weblink or an attachment for the sources you reference. Please select all answers that apply.

2. Health Workforce–Related Goals

These goals focus on safeguarding the well-being of health workers, ensuring their job satisfaction, and creating a supportive work environment that enables them to thrive in their roles. Countries use staffing level policies to directly improve the situation of health workers, typically aim to:

Protecting the health and well-being of health workers: adequate staffing level are key to ensure that health workers are not exposed to constant work overload affecting their health.

Improving retention by improving working conditions: adequate staffing levels are an instrument to retain health workers in healthcare organisations.

3. Societal Goals

These goals are outside the direct scope of the health system but are still closely linked to staffing levels. For example, the environmental impact of health workers' green skills can contribute to sustainability. A well-staffed health workforce can also positively affect economic development by supporting a healthier population, which in turn contributes to a stronger economy. Secure staffing levels during times of economic crisis have been shown to help maintain economic stability. Additionally, staffing levels can affect social cohesion, especially in areas with limited access to care, such as medical deserts or deprived urban areas.

Social cohesion: to achieve collective commitment to unity, mutual support, and shared responsibility among individuals and communities. This goal encompasses eliminating discrimination, reducing income inequality, and ensuring equal access to healthcare.

Economic development: is a vital societal goal which enables higher living standards, job creation, and increased innovation.

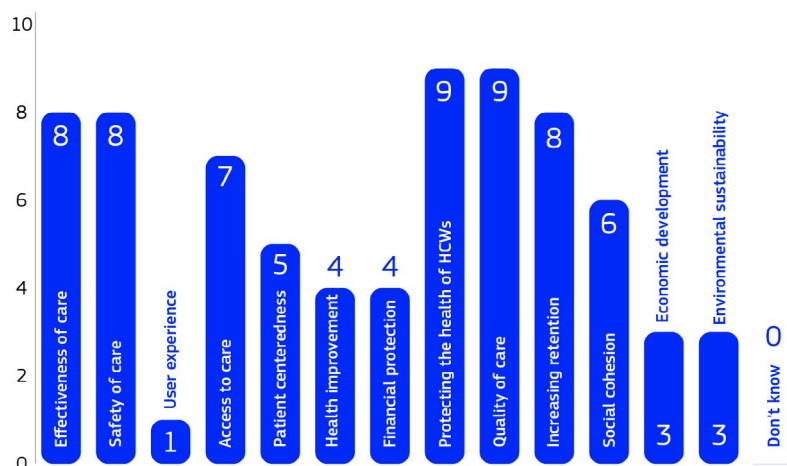
Environmental sustainability: refers to the interconnectedness of environmental concerns with broader sustainability objectives. It refers to a balanced and inclusive approach to economic growth that seeks to meet the needs of present and future generations without compromising natural resources and global ecosystems.

Note on “Intermediate Objectives”: These objectives are directly related to how well health systems deliver services and achieve overall health outcomes. They include effectiveness, safety, user experience, and access to care. Achieving these objectives is crucial for high-quality service delivery. For this to happen, health systems need the right number of skilled workers in the right places at the right times. These intermediate objectives are a key link between staffing levels and the final health system goals. If intermediate objectives aren't met, it's unlikely the system will perform well in terms of people-centered care, health improvement, and financial protection. The relationship between staffing levels and health system outcomes can be understood through these intermediate objectives.

Figure 4 below shows the answers countries, who define staffing levels at the system level, gave. As can be seen, all countries (9) indicated that quality of care and protecting the health of healthcare workers (HCWs) were the most important reasons for defining staffing levels at the system level. This hints to a strong focus on maintaining high service standards and safeguarding the workforce. Closely behind, effectiveness of care, safety of care (which can be seen as sub-components of quality), and increasing retention (each scoring at 8), suggesting that countries see staffing levels as a key driver of both patient outcomes and workforce stability.

On the other hand, user experience (1) was identified as less important for countries, indicating that it is not a primary driver in workforce staffing decisions at the system level. Similarly, economic development and environmental sustainability score 3 each, implying that they do not appear as high-priority considerations in this context. No respondents selected “Don’t know”, which potentially indicates that countries’ staffing levels at the system level are selected based on a clear rationale and linked carefully to health system objectives.

Figure 4 Reasons for defining staffing levels at the system level



Note: The 9 countries that define staffing levels at a system level responded to this question.

Figure 4 reflects a trend where immediate healthcare quality, workforce protection, and operational effectiveness seem to outweigh longer-term or indirect societal and environmental factors in staffing decisions. The results indicate that the nine countries in our sample that set staffing levels at the system level are aiming primarily to ensure safe, accessible, and high-quality care and a sustainable, motivated workforce, while also contributing to broader societal cohesion.

Sectoral level

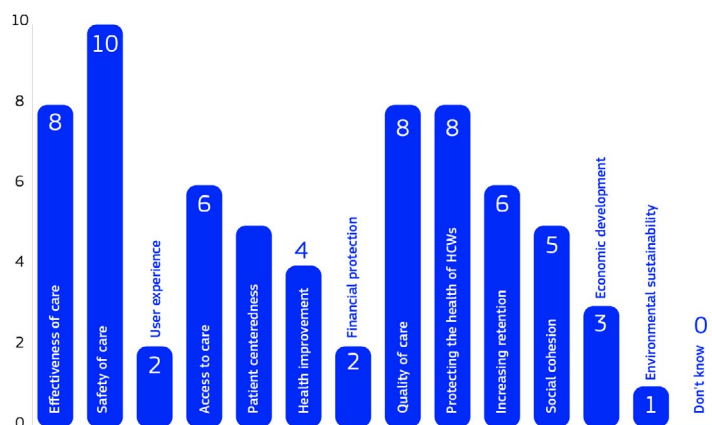
Hospitals

The 12 countries⁽⁷⁾ that filled out the questionnaire for hospitals and define staffing levels in this sector reported that safety of care (10) was the most important reason for defining staffing levels in this sector (see Figure 5). This suggests that ensuring patient safety is the primary driver of staffing decisions in hospital settings. Ensuring patient safety is followed closely by effectiveness of care (8), quality of care (8), and protecting the health of healthcare workers (HCWs) (8), indicating that both patient outcomes and staff well-being are of equal concern to these countries.

In contrast, the categories access to care (6), patient centeredness (5), health improvement (4), and economic development (3) rank lower, indicating these would not be the leading drivers for staffing decisions within hospitals across the region. User experience (2), financial protection (2) and environmental sustainability (1) are among the least reported factors, suggesting that wider societal or ecological considerations currently play a minimal role in shaping hospital-level staffing strategies.

(7) Spain defines staffing levels at the hospital sector, filled out the questionnaire, but did not fill out this question and thus the statistics only reflect 11 countries. Finland filled out the questionnaire but does not define staffing levels in the hospital sector.

Figure 5 Reasons for defining staffing levels at the sectoral level for hospitals



Note: 12 countries are included in the sample.

Overall, the data reveals that hospital-level staffing decisions are predominately shaped by the need to maintain safety, effectiveness, and quality, with a secondary emphasis on workforce protection. Broader economic, environmental, and experiential factors seem to play a smaller role.

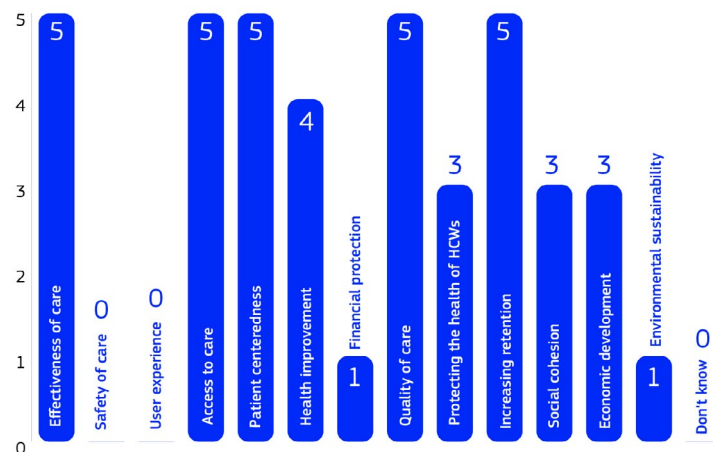
Primary care

Across the five countries ⁽⁸⁾ that filled out the sectoral questionnaire and define staffing levels in this sector, the reasons for defining staffing levels at the level of the primary care sector are multi-faceted (Figure 6). Several factors share the highest score of 5, including effectiveness of care, patient-centeredness, access to care, quality of care, and increasing retention. This suggests that, in primary care settings, staffing decisions are shaped by a balance between ensuring service effectiveness, meeting patient needs, maintaining access, delivering quality, and keeping staff in the workforce.

Safety of care (and user experience) received a score of 0, in contrast to its high ranking in hospital-level priorities. This difference for primary care may reflect the different risk profiles in primary versus acute care settings. Compared with the results for hospitals, the overall pattern for primary care suggests that primary care workforce planning is more evenly distributed across access, patient focus, and service quality, while acute safety concerns and broader environmental or economic considerations are deemed less relevant.

(8) Norway, Italy and Finland also filled out questionnaires for primary care, but do not define staffing levels in this sector and thus were not included in this part of the analysis. . In total, there are therefore only 5 countries that define staffing levels for primary care.

Figure 6 Reasons for defining staffing levels at the sectoral level for primary care



Note: 5 countries are included in the sample.

Other sectors

The other sectors for which survey responses were received were ambulatory care (3 responses), specialist care (1 response), and dental care (1 response)⁽⁹⁾. Given the limited responses in these sectors only a basic comparative analysis was possible.

When comparing hospitals and primary care to other sectors such as ambulatory, specialist, and dental care, a clear difference emerges in the underlying reasons for defining staffing levels. As previously mentioned, in hospitals the emphasis is very much on safety and quality of care, followed by protecting healthcare workers, which reflects the high-risk, acute nature of hospital services. In primary care, the priorities are more balanced across effectiveness, access, quality, and patient-centredness, showing its role in ensuring broad coverage and continuity of care rather than focusing solely on acute safety. By contrast, countries survey responses for ambulatory and specialist care cited that they adopt models that can best respond to patient demand, operating hours, and legal obligations, with goals tied to effectiveness, access, retention, and social cohesion. This could demonstrate a need for adaptability in community-based and outpatient settings, where demand can fluctuate significantly.

Overall, hospitals focus on safety, primary care on equity and patient access, while ambulatory, and dental sectors prioritise flexibility and local adaptation to ensure adequate coverage.

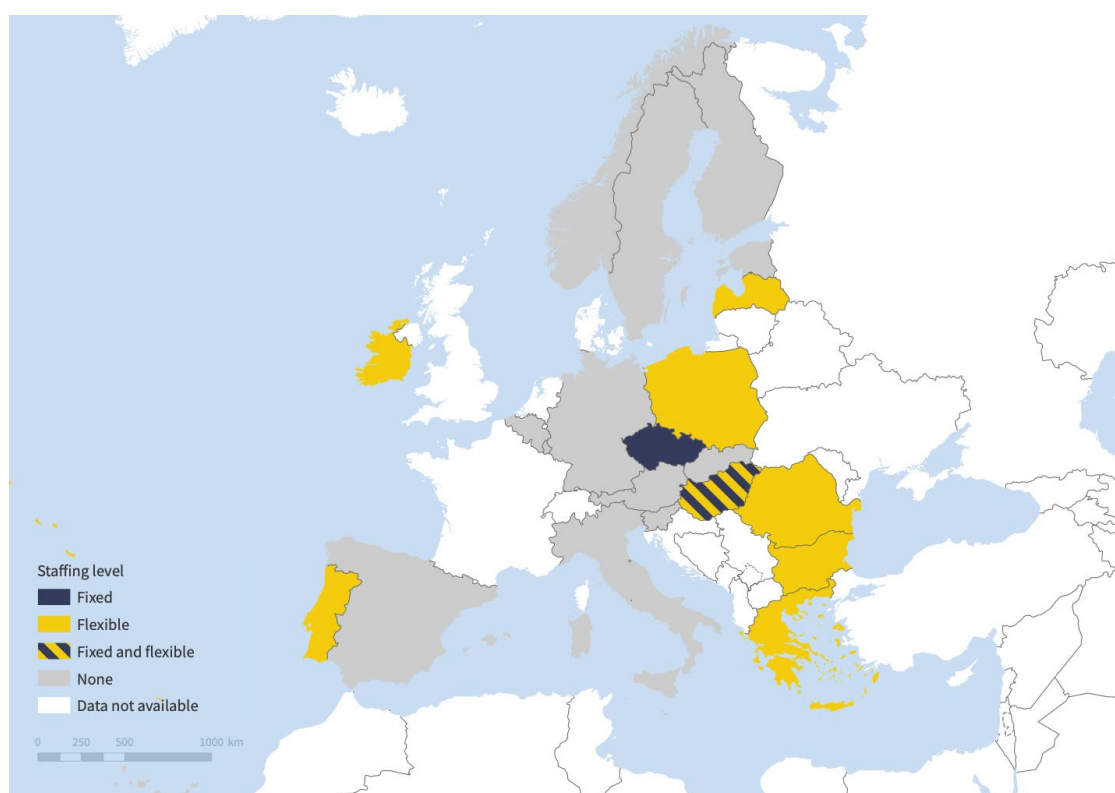
(9) One country, Austria, filled out a sectoral questionnaire for long term care, but the sector was not in the scope of the study.

Chapter 3: What kinds of defined staffing level solutions are in place?

System level

Among the 22 countries surveyed, 9 of them (Bulgaria, Latvia, Hungary, Czechia, Poland, Ireland, Portugal, Greece and Romania) report having formal system-level staffing levels in place. There is a mix of approaches used across countries including fixed, flexible and hybrid staffing level models (Figure 7).

Figure 7 Staffing level solutions in place at the system level by country



Fixed ratios are used in Czechia, with replies indicating this is due to their simplicity and measurability or for goal tracking against international benchmarks (See Box 3). Flexible approaches, such as in Bulgaria, Poland and Latvia, align workforce numbers to demographic shifts, socio-economic factors, system capacity and real-time demand (e.g. bed adjustments and service hours). Greece cites that flexible staffing levels allow for easier adaptation to local demand and evolving service needs, while also helping to mitigate staffing gaps in underserved areas. In Romania, they emphasise consensus-driven, expert-informed methods and in Ireland, the flexible approach (known as ‘the HSE (Health Service Executive) pay and numbers strategy’ - See Box 4), aims to ensure sustainable growth in HCWs employment which aligns with population needs, while enabling better pay bill control, visibility, and governance.

Hungary, on the other hand, uses a combination of fixed and flexible staffing models. This ensures that healthcare workforce levels are both legally compliant and responsive to changing societal goals.

The majority, 13 countries, do not define staffing levels at the system level. Instead, they choose to either i) delegate staffing decisions to regional authorities, individual providers, or sector-specific bodies; ii) rely on forecasting models aimed at achieving long-term workforce balance rather than fixed staffing mandates; or iii) maintain only partial arrangements tied to specific professions or service areas. Moreover, 3 of those 13 countries (Austria, Germany and Belgium) are federal systems, meaning that decision-making powers are shared between the federal government and the regions. Thus, staffing levels are often decided at the regional/federated level (Austria and Belgium). Similarly, five countries have highly decentralised health systems thereby making national staffing levels impractical or unnecessary. Spain, Italy, Sweden, Finland and Norway rely on regional autonomy, thus letting regions or autonomous communities set their own benchmarks. Examples from these two scenarios include Austria and Spain, who assign staffing responsibilities to self-governing bodies or autonomous communities. Sweden and Finland emphasise local self-governance and micro-level decision-making, and similarly, Norway states that local providers are best suited to determine staffing based on service needs and population diversity (See Box 6).

Some countries, Germany, Slovakia and Estonia, define staffing levels only for certain sectors. In Germany, staffing levels are defined for the hospital sector (Box 23). In Slovakia and Estonia, staffing levels are defined for primary care. A few countries prioritise forecasting and planning as opposed to having staffing levels. Slovakia, for example, focuses on planning regional networks and regulating training/coverage (esp. in primary and specialised ambulatory care), but has no system-level thresholds (Box 24), while Finland emphasises planning and governance by wellbeing in counties and regulates training and access with only a few recommendations and no system-level ratios (Box 7). Italy uses a national stock-and-flow forecasting model (State–Regions process) to plan future training places and workforce balance, but not to define current staffing levels (Box 13).

The final reason that countries do not define staffing levels is a lack of evidence or fear of service disruption. Slovenia cited insufficient evidence to justify establishing staffing levels (Box 26) while Sweden found that fixed staffing levels might disrupt efficiency (Box 8). Malta also cited similar reasons to Sweden but added that the Public Health Service authorities were cautious of introducing official staffing levels, as they feared such an action might interrupt service provision due to challenges related to inadequate supply and possible industrial unrest (Box 5).

Together, these explanations highlight a common tension between the predictability of fixed ratios and the flexibility needed to adapt to local contexts.

BOX 3 - Czechia uses fixed staffing levels to achieve clear, quantifiable targets for health professionals

How the model works

Czechia employs a fixed staffing model anchored in its national health strategy Health 2030. The central target is set at 32 health professionals per 1,000 inhabitants by 2030, covering all major professional groups such as doctors, nurses, and other healthcare staff. This aspirational goal is complemented by a detailed personnel directive that lays out staffing requirements for different parts of the health system. For outpatient care, the directive requires one doctor per specialty plus one nurse, while in primary care, similar rules apply with additional standards for group practices, which must have between one and three general practitioners. Hospitals are also covered by sector-specific rules under the same directive.

Why they use a fixed model

The fixed ratio approach in Czechia is driven by several factors. Firstly, clarity and simplicity: fixed ratios are straightforward to measure, track, and communicate, both to policymakers and to the public. This makes them politically and administratively practical. Secondly, they support comparability across time and with other countries, as the aspirational target was partly shaped by international benchmarking. Thirdly, fixed ratios allow for easier compliance monitoring, since authorities such as health insurance funds, ministries, and regional bodies can quickly check whether staffing meets defined thresholds.

Goals behind the model

Czechia's staffing ratios are designed to serve multiple objectives. Health system goals are mostly access and quality, while safety is the main goal in hospitals. Staffing ratios aim to reach those by ensuring that the workforce is adequate and evenly distributed. At the workforce level, the model supports retention and protection of healthcare workers, preventing overload by guaranteeing minimum staffing. Finally, it also provides an aspirational benchmark to align future workforce planning with population needs.

Summary

Czechia's fixed ratio staffing model combines a clear national target (32 per 1,000) with detailed operational rules for specific care settings. Its main strengths lie in simplicity, measurability, and comparability, making it easier to enforce and communicate. The model prioritises both patient safety and workforce sustainability, while serving as a long-term roadmap for achieving adequate staffing by 2030.

BOX 4 - Ireland uses flexible staffing to align the health workforce with system needs

How the model works

Ireland uses a flexible model for defining staffing levels at the system level, managed primarily through the Health Service Executive (HSE) Pay and Numbers Strategy (PNS). This strategy is an annual publication derived from the national Budget and the HSE's National Service Plan. It sets out the number and allocation of staff that the HSE, the national public health provider, has approval and funding for in a given year. Staffing ceilings are defined in Whole Time Equivalent (WTE) terms, which provide a standardised way to account for staff numbers across different professions.

The process involves several actors. Staffing ceilings are agreed between the Department of Health (D/Health) and the Department of Public Expenditure, National Development Plan Delivery and Reform (DPENDPDR). Once the overall ceiling is set, responsibility for deciding which posts will be filled lies jointly with D/Health and the HSE. In addition, the Department of Children, Disability, and Equality (DCDE) is responsible for funding staff working in disability services, reflecting the system's integration of health and social care.

Why they use a flexible model

Ireland adopted this approach to strengthen pay bill management, improve oversight, and increase governance transparency. Unlike rigid fixed staffing ratios, the PNS allows the system to adapt workforce numbers annually based on budgetary realities, service needs, and evolving health priorities. It provides the government with a high degree of control over expenditure while enabling the HSE to align staff numbers with operational and population health needs.

Goals behind the model

This model aims to achieve a balance between health system performance and workforce well-being. At the system level, the objectives include effectiveness of care, safety, quality, patient-centredness, and financial protection. At the workforce level, the strategy seeks to protect staff well-being and improve retention by ensuring staffing numbers can grow sustainably in line with demand. Importantly, the approach also emphasises skill-mix considerations, recognising that workforce planning is not just about numbers but also about having the right combinations of professionals to deliver care effectively.

Summary

Ireland's flexible staffing model prioritises financial sustainability and adaptability. By linking staffing levels directly to budgetary processes and service planning, it ensures that workforce growth is both affordable and aligned with population needs. This model avoids the rigidity of fixed ratios, instead focusing on governance, oversight, and strategic workforce planning that integrates skill-mix, staff well-being, and health system objectives.

BOX 5 - Malta's evidence-based workforce planning approach

Malta has chosen not to introduce binding staffing ratios or formal minimum thresholds. Authorities in the Public Health Service were concerned that setting such requirements could disrupt service provision due to limited workforce supply and create risks of industrial unrest, given the highly unionised nature of the health sector. Instead, staffing decisions are made based on need, allowing flexibility and responsiveness to immediate service demands.

What it does instead

Rather than relying on fixed ratios, Malta has embraced a data-driven, evidence-based workforce planning approach. The shift began with WHO-supported consultations in 2019–2020, which laid the groundwork for sustainable health workforce forecasting. In 2022, the Ministry for Health and Active Ageing launched the Health Workforce Strategy (2022–2030), aligning national efforts with EU Joint Action HEROES and WHO frameworks (i.e. the WHO Global Strategy on Human Resources for Health: Workforce 2030 and the WHO National Health Workforce Accounts (NHWA) framework).

A central feature of this strategy is the adoption of the Nivel forecasting tool (Netherlands), introduced with WHO's support under the Recovery and Resilience Plan. This tool enables systematic workforce projections tailored to Malta's healthcare context. It has spurred critical discussions on training expansion, task shifting, and resource allocation to address shortages.

In line with these changes, Malta is also building a centralised data repository for healthcare professionals, integrating information from:

- Regulatory bodies
- Jobs Plus (employment agency)
- Identita' Malta (identity and immigration agency)
- Educational institutions

By linking datasets with unique identifiers, Malta ensures that workforce intelligence is accurate, timely, and comprehensive.

Summary

The Maltese approach combines international collaboration (EU and WHO), advanced tools, and stakeholder engagement to ensure that staffing policies remain both evidence-based and adaptable to changing healthcare needs.

Box 6 - Norway opts for quality over quotas

Norway does not mandate specific staffing ratios across its health and care system. Instead, it uses a quality-based governance model where each service provider must ensure their care meets the defined standard of “soundness.” This allows:

- Local discretion in staffing decisions.
- Adaptation to patient complexity and service type.
- Avoidance of rigid staffing formulas that may not suit all contexts.

This very flexible approach to staffing levels in Norway works well and is necessary for several reasons. First, hospitals and care providers in Norway vary widely in the populations that are served (densely populated cities like Oslo versus remote villages). In addition, patient needs, i.e. acute care versus long-term care also vary. In sum, it is believed that local providers know best how to staff appropriately for their given situation and circumstances. Secondly, Norway’s healthcare system is decentralised. The country has 357 municipalities for primary care that range from tiny villages with a population no greater than 200, to large metropolitan cities. This type of decentralisation requires a certain amount of flexibility for continuous and effective care across the very diverse settings. Thirdly, Norway already anticipates future workforce shortages and thus wants to keep the flexibility and agility of its workforce organisation in order to more easily be able to task shift or restructure roles, thereby better being able to adjust teams dynamically based on real-time needs.

Summary

Norway deliberately chooses not to set staffing levels and this decision is rooted in its decentralised health system, geographic diversity, and desire for future adaptability. By empowering local providers and focusing on service quality rather than numbers, Norway seeks to maintain high standards while adapting to demographic and workforce challenges.

BOX 7 - Flexibility over ratios: Finland’s alternative to staffing levels

The Finnish approach is rooted in its governance model: health and social services are the responsibility of self-governing wellbeing counties, which hold autonomy in organising care. Because of this decentralisation, rigid national staffing rules are considered incompatible with local flexibility.

Recommendations exist with one clear minimum staffing rule:

Although there are no national staffing levels, recommendations exist in certain sectors of public health:

- Child health clinics: 1 public health nurse per 320 children, and 1 physician per 2,270 children.
- School health services: 1 public health nurse per 460 pupils, and 1 school doctor per 2,100 pupils.
- These recommendations, published by the Finnish Institute for Health and Welfare (THL), are advisory, not mandatory, and mainly serve as guidance for wellbeing counties.

In the context of social care, specifically elderly care (formally part of social services rather than healthcare) Finland has introduced one clear minimum staffing level:

- 24-hour care units: at least 0.6 employees per client.

This reflects concerns about ensuring safe and dignified care for older people, where staffing shortages have been politically and socially sensitive.

Why are there no minimum thresholds in healthcare?

Successive Finnish governments have chosen not to legislate minimum thresholds for doctors and nurses. Instead, they rely on alternative policy tools to safeguard access and quality:

- Strengthening the governance and accountability of wellbeing counties.
- Encouraging workforce planning and continuity of care through service integration.
- Focusing on monitoring outcomes and access rather than prescribing inputs like staff numbers.

The rationale is that fixed ratios may not reflect regional realities, workforce availability, or evolving service needs.

Summary

Although Finland has resisted introducing universal staffing levels, debates continue, particularly in primary care, about whether continuity of treatment could benefit from more explicit benchmarks. Currently, however, the Finnish model prioritises flexibility, decentralised responsibility, and outcome-based monitoring over rigid national ratios.

BOX 8 - Sweden's alternative to staffing levels

Sweden does not apply minimum staffing thresholds for healthcare professions. Staffing decisions are instead made at the regional level, where 21 self-governing regions are responsible for allocating budgets and managing licensed healthcare personnel. Within these regions, hospitals and primary care organisations have the autonomy to organise their staffing models based on local needs and contextual challenges, rather than adhering to rigid ratios.

Since 2022, the National Board of Health and Welfare has introduced reference values for doctors in primary healthcare:

- 1,100 inhabitants per specialist physician
- 550 inhabitants per physician under training

These values are non-binding but provide guidance to regions for planning. The government has signalled that it may extend this reference value approach to other healthcare professions, suggesting a flexible alternative to mandatory ratios.

Instead of national thresholds, Sweden relies on a combination of:

- Regional autonomy to define staffing models wherein staffing decisions are delegated to regions, hospitals, and primary care centres to adapt to local conditions.
- Reference values to guide planning without imposing rigid rules.
- Task shifting and multidisciplinary teams to maximise efficiency.
- Local decision-making at the micro-level, where hospitals and care organisations determine what staffing best fits their context.

Summary

Sweden's model reflects a deliberate choice to avoid fixed staffing levels. By combining reference values, decentralised governance, and flexible team-based approaches, Sweden ensures that staffing decisions are context-specific, adaptive, and evidence-informed. This system prioritises responsiveness and efficiency over national one-size-fits-all standards.

Sectoral level

Approach to staffing levels in hospitals

Out of the 22 countries surveyed, 12 (more than half) responded to the hospital survey and all 12 countries define staffing levels in hospitals. Nine of the 12 countries adopt fixed staffing ratios for at least one of the core clinical roles, i.e. doctors and nurses, across most unit types (general wards, ICU, ER, surgical, maternity, psychiatric and “other” areas, i.e. physiotherapy, midwifery, psychology) (Figure 3). Slovakia, Czechia, Germany and Austria (Box 9) uniformly mandate fixed doctor-to-bed and/or nurse-to-bed minima, often tied to bed counts or specialty-specific quotas and they are generally embedded in laws or budget agreements. For instance, Slovakia sets maximum patient loads per nurse, doctor, and other staff by department. As previously mentioned, the appeal of fixed ratios lies in its ability to provide predictable staffing standards and to reassure both staff and patients regarding minimum care guarantees.

Alongside these fixed models, some countries blend in flexible approaches to adapt to fluctuating needs (Figure 3). Hungary and Portugal, for example, use capacity regulations, referral-network guidelines, or socio-demographic calibration to scale staffing in response to demand, population needs, or service complexity. Hungary’s dual framework combines minimum thresholds per bed with flexible capacity planning (Box 20), while Portugal supplements fixed ratios for nurses with flexible referral-network guidance for specialist services (Box 12). Luxembourg uniformly applies a fixed, activity-based ratio for doctors (e.g. doctor-to-delivery), while adopting a more flexible approach to nurse-to-patient ratios in hospital wards. (Box 10).

Bulgaria, Spain and Greece stand out as they all use an exclusively flexible model to define staffing levels in hospitals. Bulgaria uses a flexible calibration system anchored in the National Health Card, Ministry of Health standards, and framework agreements with the National Health Insurance Fund. Here staffing levels are not fixed but adjusted to match demographic realities, workforce availability, and the evolving needs of the population (Box 14). Spain’s hospital workforce planning is primarily handled by the autonomous communities, with staffing levels adjusted locally based on service demand, population needs, and available workforce supply rather than nationally mandated ratios. This decentralised and adaptive approach reflects Spain’s devolved healthcare system, where flexibility is essential to address regional differences in hospital demand and resource capacity (Box 11). In Greece, staffing is determined annually through a multistep process designed to remain flexible in the face of institutional and geographic constraints. Norway is distinct in that it relies on a qualitative “soundness requirement” rather than numerical ratios (See Box 6).

More broadly, countries define hospital staffing along a spectrum: many use legally mandated ratios for doctors and nurses, often linked to bed numbers or unit types to ensure predictability and minimum standards of care, while others adopt more flexible, demand-driven and data-informed approaches. Together, these models illustrate how health systems balance regulatory certainty with adaptability in hospital workforce planning.

BOX 9 - Austria uses layered legal and planning tools to define hospital staffing levels

Austria regulates hospital staffing through a layered system of laws, planning frameworks, and funding rules, which together create mostly fixed staffing ratios for doctors, nurses, and other professionals. The system rests on three main instruments:

Austrian Structural Plan for Healthcare (ÖSG) – a national planning framework that provides expert legal guidance. Although much of the ÖSG content is not legally binding, hospitals can be held accountable if they fail to comply. The ÖSG stresses that staffing must be sufficient to meet diagnostic, therapeutic, and nursing needs, and that high-quality care requires all relevant professional groups to be adequately represented.

Federal Hospital Act (KAKuG) – the legal foundation. It obliges hospitals to calculate staffing needs annually by professional group, department, and unit. These staffing plans must be overseen by qualified staff and submitted to the state government, embedding staffing into routine hospital governance.

Austrian DRG System (LKF) – the financial instrument. It links reimbursement to quality criteria that include minimum staffing thresholds. For example, ICUs, stroke units, psychiatry, and palliative care must meet specific full-time equivalent (FTE) staffing ratios per bed. Hospitals that fall below these thresholds risk losing full DRG-based reimbursement. While the rules are national, federal states can adapt resource allocation for hospitals with special care functions.

How it works in practice and why they use it

This multi-tiered system ensures that hospitals operate under clear staffing expectations while still allowing states (Länder) some flexibility. The ÖSG provides strategic guidance, KAKuG enforces legal accountability, and the LKF ensures compliance through financial incentives.

This model reflects the need to safeguard effectiveness, patient safety, equitable access, staff well-being, and quality of care. Fixed ratios provide clarity and accountability, while state-level flexibility recognises Austria's federal structure and the varied contexts across regions.

Because Austria is a federal system, uniform implementation is difficult. Staffing ratios may be interpreted and applied unevenly across states, with negotiations shaping how the rules work in practice.

Summary

Austria's hospital staffing model combines planning, law, and funding to enforce minimum staffing levels across critical hospital units. It strikes a balance between national standards and regional flexibility, ensuring patient safety and quality while adapting to local needs.

BOX 10 - Balancing fixed rules with budget-linked flexibility in Luxembourgish hospital staffing

How it works

Luxembourg regulates hospital staffing through a hybrid system that combines fixed standards with budget-linked negotiations. The foundation is the Hospital Law of 2018, which defines hospital departments, essential standards, and minimum requirements for certain specialties such as obstetrics. Annex 2 of the law lists the health professions covered—ranging from doctors and nurses to pharmacists, physiotherapists, dietitians, social workers, and nursing assistants. Additional regulations, such as the 2019 decree on emergency departments, set detailed operational and staffing requirements. Hospitals are authorised for specific services and bed numbers, which form the baseline for staffing calculations.

Beyond legislation, staffing levels are shaped by the biennial budgeting process under the Social Security Code. Every two years, the government sets a global hospital budget. Staffing levels are then negotiated between the National Health Fund (CNS) and the Federation of Luxembourgish Hospitals (FHL). A joint Staffing Standards Commission calculates the required full-time equivalents (FTEs) per staff category for each hospital unit, based on past activity data and projected needs. This ensures predictable financing and consistency across hospitals nationwide.

Fixed and flexible elements

During each two-year cycle, many staffing ratios are effectively fixed, particularly for units like geriatrics, dialysis, and hospital pharmacies. At the same time, there is flexibility for other departments, where staffing can be adjusted annually using measurement tools linked to patient needs and service activity. A more advanced measurement tool is planned for 2026 to refine these calculations and improve adaptability. Negotiations also allow for adjustments based on organisational innovations, resource sharing, or new technologies.

Why they use it

Luxembourg's hybrid model provides both stability and adaptability. Fixed ratios give hospitals clarity and financial security, while flexible mechanisms allow the system to respond to changing demand and evolving care complexity. This dual approach supports quality, patient safety, and workforce sustainability, while maintaining financial control through the national budget.

Summary

Luxembourg's hospital staffing system balances nationally standardised rules with negotiated flexibility. It secures predictable funding, ensures equity across hospitals, and creates space for local adjustments. By combining law, budgeting, and measurement tools, Luxembourg aligns staffing governance with its goals of effectiveness, safety, financial protection, and long-term workforce sustainability.

BOX 11 - Flexibility and decentralisation drive Spain's hospital workforce management plans

How it works

Spain uses a flexible, decentralised model for defining hospital staffing levels, with responsibilities delegated to the Autonomous Communities (AACC) and the autonomous cities of Ceuta and Melilla. Under the Framework Statute (Law 55/2003), each region must develop and update Human Resources Management Plans. Hospitals also create their own staff planning models within regional frameworks. Importantly, there are no binding national minimum thresholds, except during strikes, when the administrations must set minimum service levels to protect both patient care and the right to strike.

The role of the regions

Regional authorities apply tools such as population profiling, workload analysis, and workforce projections, in order to adapt methods to local needs. This allows for increased flexibility but creates significant variation across communities. At the national level, the Ministry of Health provides the strategic coordination by publishing reports on professional needs (e.g., *Oferta y necesidad de médicos especialistas en España 2023–2035*, *Necesidades Profesionales de Enfermería 2024*). These serve as common reference points for regional planning.

Coordination is further supported by the NHS Human Resources Commission, which brings together the Ministry of Health, the AACC, and other ministries such as Education, Universities, Defence, and Justice. This forum strengthens the link between staffing evaluations, training capacities, and future workforce needs. Investment and recruitment are funded through regional budgets, while national and regional regulations connect staffing levels to training quotas for students and medical residencies.

Summary

Spain's flexible and decentralised approach for defining hospital staffing levels provides adaptability and regional ownership, but the lack of binding national thresholds has the potential of leading to uneven workforce distribution and access across the regions.

Approach to staffing levels in primary care

Five countries report having defined staffing levels for primary care (Austria, Bulgaria, Hungary, Portugal, and Spain)⁽¹⁰⁾. These countries are split between fixed and flexible models across doctors, nurses, and other professionals. Bulgaria, for example, fixes nurse coverage, but allows for flexible doctor and some nurse staffing based on the National Health Card, clinical standards and collective-bargaining agreements. Austria sets fixed ratios for doctors and nurses yet applies flexible guidelines for allied professionals within its regional planning frameworks (Box 33). Portugal fixes nursing-staff requirements per Family Health Unit for human resource planning (Box 12). Hungary uses a mix of fixed and flexible staffing methods in primary care to address the diverse needs of the population. Fixed staffing for all doctors, nurses and health visitor nurses supports continuity of care, while flexible staffing for nurses allows them to adapt their roles, such as providing home-based care or supplementing GP services, especially in underserved areas. Spain uses a flexible model that is administered by the regions and autonomous communities as they see fit.

Countries that set fixed ratios aim to guarantee baseline safety, equity, and predictability, thus ensuring every patient list or care unit meets a minimum doctor or nurse threshold and linking these minima to funding, licensing, or accreditation. The flexible models were chosen because they accommodate demographic shifts, case complexity, and service demand, allowing countries like Bulgaria, Hungary, Portugal, and Austria to scale staffing with age-weighted user lists or workload forecasts.

Norway does not define staffing levels for primary care but instead, as for hospital care, requires municipalities to satisfy a broad “soundness requirement” that ensures key professional roles remain available, thus preserving local autonomy (Box 6).

Three countries (Czechia, Italy, and Poland) did not fill out a sectoral level questionnaire for primary care, rather they added this information into their system level questionnaire. Czechia, for example, sets a clear national target of 32 health professionals per 1,000 inhabitants by 2030 and complements this with directives such as requiring one GP and one nurse per practice. These rules, shaped through expert consultations, are designed to be easy to measure, enforce, and communicate while supporting access, safety, and workforce retention. In Poland, there is no single staffing standard for primary care, but a Regulation recommends that the optimal number of patients assigned per primary health care doctor should not exceed 2,500 patients. Poland uses also fixed standards based on expert consensus and international benchmarking, alongside flexible workforce planning and demand-supply forecasting models that anticipate regional needs and future physician availability. This ensures both immediate coverage and long-term sustainability of general practice. Italy, by contrast, does not apply

(10) Italy reported on primary care/community care but does not have staffing levels in place at this time because the system is undergoing a major reorganisation and therefore finds itself in a “transitional phase”. All staffing models are still being developed, are currently flexible and are tied to ongoing reforms. Full implementation and the potential for standardisation is expected by 2026, once the Recovery and Resilience-funded (RRP) restructuring is complete.

minimum thresholds in primary care but instead relies on a sophisticated forecasting methodology, developed under the Joint Action HEROES, that projects supply and demand of health professionals 20–25 years ahead. This model informs enrolment in health degree programmes and aims to prevent imbalances that could undermine the future capacity of primary care. Slovakia also did not fill out a separate questionnaire for primary care but added this into ambulatory care where they mandate a fixed doctor-per-patient minimum.

BOX 12 - Portugal balances fixed nursing ratios with flexible team-based allocation

How it works

Portugal's primary care staffing is centred on the Family Health Unit (USF) model, introduced in 2005 as part of a reform to modernise care and expand universal access. Under Decree-Law No. 298/2007, updated by Decree-Law No. 103/2023, USFs were granted greater organisational and financial autonomy. Each unit is staffed by multidisciplinary teams of doctors, nurses, and administrative staff working collaboratively to deliver patient-centred care.

The law also introduced minimum patient list sizes for doctors and nurses, calculated through “weighted units” that adjust for sociodemographic factors such as age and gender. These lists are reviewed twice a year and updated whenever staffing changes occur, ensuring workforce capacity matches demand. For nurses, fixed staffing ratios are defined through the Safe Nursing Care Staffing Standards. For doctors, the approach is more flexible, allowing team size and composition to adapt to the needs of registered populations.

Complementary units

Alongside USFs, two additional structures support primary care:

- Community Care Units (UCCs): These consider local demographic and social factors, generally applying a ratio of one nurse per 5,000 inhabitants (preferably a specialist nurse).
- Public Health Units: These follow population-based standards such as one doctor per 25,000 inhabitants and one nurse per 30,000 inhabitants. These ratios are non-binding and revised annually to reflect changing needs.

Why they use it

Portugal's hybrid approach is designed to ensure both safety and adaptability. Fixed minimum nursing standards safeguard patient safety, while the flexible allocation of doctors and team structures allows the system to respond to local demand, population dynamics, and evolving health priorities.

Summary

By anchoring standards where they matter most and leaving space for adaptation elsewhere, Portugal favours better adaptation to its context, thereby being able to influence both the quality and the efficiency of its system.

Approach to staffing levels in other sectors (ambulatory, specialist and dental care)

Only 4 of the 22 countries filled in questionnaires on **ambulatory care** (Slovakia, Hungary, Finland and Austria), with only Hungary and Slovakia indicating that they define staffing levels for this sector. Slovakia uses doctor-to-patient ratios that scale with age groups and patient volumes, making them simultaneously predictable and adaptable. Hungary applies a dual system: fixed minimum thresholds for staffing per specialist hours, combined with flexible capacity regulations revised every three years based on population demand and utilisation data. Austria, by contrast, does not define staffing ratios in ambulatory care, relying instead on collective agreements between insurers and professional associations.

In **dental care**, Norway (the only country to submit a response to this sector) applies a flexible, needs-based approach. Under the Dental Health Services Act, counties must ensure adequate provision, particularly paediatric dental specialists, but no fixed ratios exist nationally. Instead, capacity planning is demand-driven, supported by state incentives to create specialist positions.

Defining staffing levels in other sectors is generally more fragmented, with countries like Slovakia and Hungary applying mixed fixed-and-flexible approaches in ambulatory care, while Austria and Norway rely instead on agreements or broad “soundness” requirements. In dental care, regulation is demand-driven without fixed ratios (Norway), reflecting a preference for flexibility over uniform national standards.

Chapter 4: How are staffing levels governed?

System level

What are the governance solutions, how are the staffing levels agreed upon and who is involved in the agreement process?

This section looks exclusively at the nine countries that define staffing levels at the system level, as the countries not defining staffing levels at the system level were asked to skip this section. Austria, Spain and Belgium are also included as they define staffing levels at the regional level given their federally structured or decentralised systems. Italy, Slovakia and Sweden are loosely included because they answered this part of the questionnaire despite not defining staffing levels.

Solutions for staffing levels are typically set or steered from a central level⁽¹¹⁾ with varying degrees of regional and professional input

Generally, the Ministry of Health takes the lead in setting or coordinating staffing standards and they are, at times, supported by other ministries (usually Finance, Education or Interior). This combination of efforts ensures that staffing levels are not only aligned with health service goals but also with broader fiscal and public sector planning constraints. In some cases, such as in Hungary (Box 20) and Greece (Box 18), the rules are found within legally binding decrees or laws, which specify minimum professional requirements for hospitals and other health institutions. Elsewhere, such as in Portugal and Ireland, the national benchmarks function as reference points rather than strict thresholds, allowing more room for local adaptation (Box 21).

In Spain, staffing governance is delegated to the autonomous communities, which plan and hire according to local population needs, while the national government produces periodic reports to guide strategic alignment (Box 19). Austria maintains a federal or national leadership role, where ministries and sickness funds shape frameworks based on evidence and financing responsibilities. Italy presents yet another model that is nationally controlled but fiscally driven. Here expenditure ceilings are tied to historic spending and determine how regions can staff their health services (Box 13). Belgium shows a federal balance: the Federal Ministry of Health is responsible for overall regulation and financing frameworks, but the federated communities (Flemish, French, and German-speaking) and regions play a strong role in workforce planning, training capacities, and the organisation of care delivery (Box 22).

A wide range of actors are involved in finding agreement on governance arrangements

In Bulgaria (Box 14) and Latvia (Box 17), professional organisations and academic institutions play a key role in setting staffing levels. These are both directly involved in drafting rules and validating analytical studies. In Romania, an Inter-institutional Decision-Making Group brings together universities, professional bodies, local authorities, and the Ministry of Health to shape staffing approaches (Box 15). In Portugal, central agencies work with the Medical Association and professional councils to define training quotas and to align staffing maps with service profiles (Box 21).

Belgium, with its federal model, requires consensus-building between federal institutions, federated communities, and professional councils, for example, negotiations over quotas for doctors and dentists are made with medical associations and universities (Box 22). In Spain,

(11) Naturally, depending on a country's institutional set up (federal or central), governance solutions for staffing levels in health systems are structured around central or regional stewardship.

staffing rules are determined regionally, with national-level coordination limited to strategic guidance (Box 19). Italy relies on binding regulations negotiated between ministries of health, finance, and other state actors. In Austria, the involvement of sickness funds and professional self-governments ⁽¹²⁾ help to ground decisions in financial realities and professional practice.

Across all systems, the inclusion of professional chambers, universities, and insurers/payers ensures that standards are grounded not only in government policy but also in professional practice and service realities.

Evidence-based tools and methods provide a shared analytical foundation for decision-making

Countries typically use population profiling, workload analysis, and workforce projections to determine the adequacy of staffing relative to demographic trends and service demand. Hungary has developed a particularly detailed set of operational indicators ranging from patient flow data to case-mix indexes and travel accessibility measures, to guide decisions (Box 20) while Latvia (Box 17) and Ireland rely on workforce modelling and needs-based projections. These methodologies lend transparency and legitimacy to the governance process by grounding decisions in measurable, reproducible criteria.

Spain's regions use similar methods, although are granted more flexibility in order to adapt their own analytical tools. Portugal uses workforce and activity-based modelling, and Austria grounds its rules in clinical guidelines and scientific literature. Italy traditionally relied on fiscal controls but is now experimenting with analytical methods to better align staffing with service needs. Belgium combines federal and community-level analyses: The federal level analyses professional activity at each level. Activity data, education data and population healthcare demand are used to produce workforce projections, which are the basis to set the federal quota. The federated communities then use these projections to allocate sub-quotas across medical specialties within their overall quota.

BOX 13 - How Italy governs health staffing

Although Italy does not define staffing levels, it manages staffing through an intricate and shared governance system that combines State-Regions agreements, long-term forecasting, and strict expenditure controls. This reflects the country's decentralised health system, where both national ministries and regional authorities play a role.

Workforce planning and forecasting

- Training needs: Each year, access to health profession degree courses (medicine, nursing, pharmacy, psychology, etc.) is defined by a State-Regions Agreement (Article 6-ter, Legislative Decree No. 502/1992). The latest agreement was signed in July 2024. Forecasting uses the “stock and flow” model developed under the EU Joint Action on Health Workforce Planning, projecting supply and demand over 20-25 years while factoring in retirements, student intake, study success, and labour market absorption.
- Specialist medical training: Every three years, Regions and Autonomous Provinces estimate their need for specialist doctors. These figures are reviewed by the Ministries of Health, Universities and Research, and Economy and Finance. A State-Regions Agreement confirms the totals, followed by an inter-ministerial decree that allocates specialist training contracts.

(12) In Austria, these are autonomous professional bodies such as medical chambers, nursing chambers, and other professional associations, that represent and regulate health professionals, rather than government authorities.

Expenditure controls

Staffing is heavily shaped by personnel spending caps in the National Health Service. Since 2019, spending has been tied to 2018 levels (or, in some cases, adjusted from 2004 levels), with only limited annual increases linked to the Regional Health Fund. A 2024 reform (Decree-Law No. 73/2024) introduced some flexibility: regions may increase spending by up to 15% of annual fund growth, provided they adopt a new methodology for identifying staffing needs. These plans must then be approved by both the Ministry of Health and Ministry of Finance.

Stakeholders involved

Governance is shared across multiple levels:

- National level: Ministry of Health, Ministry of Universities and Research, Ministry of Finance.
- Regional level: Regions and Autonomous Provinces, which draft estimates and plans.
- Professional bodies: National federations (e.g., FNOMCeO for doctors) contribute to needs assessments.
- Universities: Allocate training places after agreements are finalised.

Summary

Italy's governance of staffing does not rely on minimum ratios. Instead, it uses forecasting models to align future supply and demand, training capacity planning through State-Regions negotiations and very strict spending caps to control costs. This system aims to balance workforce sustainability and financial discipline, but rigid expenditure limits remain a major barrier to flexibility in addressing staffing needs.

BOX 14 - Stakeholders agreements on staffing levels in Bulgaria

How it works

The governance of staffing levels in Bulgaria is shaped by a negotiated and multi-stakeholder process that combines legal obligations with strong social partnership. At the centre of this arrangement is the Ministry of Health, which plays a coordinating and decision-making role. The Ministry is responsible for defining minimum medical standards and determining personnel needs in consultation with professional organisations and the National Health Insurance Fund (NHIF). These negotiations establish staffing rules that balance workforce supply, service demand, and budgetary constraints.

Stakeholder inclusion

Professional organisations, including medical and nursing associations, contribute their expertise by defining rules of good medical practice and ensuring that staffing frameworks reflect clinical standards and patient safety requirements. Their role is important in ensuring that staffing levels are not only adequate in number but also aligned with quality of care. The NHIF provides a financial perspective, as it funds healthcare providers and ensures that staffing agreements are consistent with reimbursement models. Together, these actors draft contractual agreements, such as the National Framework Agreement, which sets out staffing obligations that are legally binding for providers.

Compliance

Oversight is provided by the Executive Agency for Medical Supervision, which, along with the NHIF, monitors the compliance of the agreed upon staffing levels. Sanctions, including contract termination or administrative penalties, are possible but rarely applied. Instead, the system relies on continuous negotiation and partnership. When workforce shortages occur, the Ministry of Health acts jointly with the Ministry of Education and Science and medical universities to expand training places, adjust specialisation pathways, thereby ensuring longer-term workforce sustainability. The Council of Ministers also intervenes by decree, setting student intake quotas and allocating the necessary funding for higher education institutions.

Tripartite Agreement and multi-level governance

Financial and workforce planning is underpinned by the Tripartite Agreement between employers, unions, and the Ministry of Health, which establishes wage frameworks, income guarantees, and funding for specialisation. This reflects Bulgaria's tradition of tripartite social dialogue, where unions and employer groups actively shape workforce policies alongside government actors. Such negotiations ensure that staffing agreements are not only legally enforceable but also socially accepted by the key actors responsible for delivering and financing care.

Summary

Bulgaria's governance of staffing levels illustrates how law, negotiation, and partnership are combined to align workforce planning with training capacity, financing, and professional standards. The Ministry of Health holds the central stewardship, but outcomes depend on active cooperation with professional bodies, the NHIF, unions, universities, and other ministries. This model highlights the importance of multi-level governance in ensuring that staffing agreements are both realistic and enforceable, while also adaptable to workforce and population needs.

Is there a link between staffing levels and HWF planning, investment and training capacities?

Many EU countries are using staffing rules as strategic tools to align workforce planning and training with service needs

In many countries, staffing levels are closely linked to workforce planning frameworks. Bulgaria (Box 14), Poland (Box 16), Hungary (Box 20), and Greece use quotas for medical residencies and nursing training that are adjusted to meet thresholds set in law or regulations. Latvia adjusts medical education and residency capacities in line with workforce shortages and OECD benchmark indicators⁽¹³⁾. Similarly, Portugal explicitly links minimum or reference staffing levels to the allocation of residency places and specialist training slots. Hungary integrates staffing thresholds into training needs assessments for nurses and postgraduate quotas, ensuring that education outputs reflect service requirements. Czechia increased medical training places as part of its national workforce strategy tied to staffing targets, while Ireland uses demand and supply modelling to guide workforce reforms, ensuring alignment between staffing needs and care area projections. Romania links staffing levels to regional health masterplans that integrate planning for workforce generation and distribution (Box 15). In Austria, planning is shaped by agreements between federal and state authorities, sickness funds, and professional stakeholders, using health needs assessments and literature evidence to influence both workforce planning and service organisation. These examples demonstrate how staffing rules can become planning tools that anticipate and manage future workforce gaps.

(13) OECD benchmark indicators refer to a set of internationally comparable metrics used to assess the size, composition, distribution, and sustainability of the health workforce across countries. They aim to allow countries to identify gaps relative to peers and to inform workforce planning and education policies (see OECD, 2016 for examples).

Staffing levels also serve as a lever for investment decisions in some countries

In Italy, expenditure ceilings legally tie staffing to fiscal planning, ensuring that recruitment aligns with budget constraints set jointly by the Ministry of Finance and Ministry of Health (Box 13). Poland (Box 16) links adequate employment levels with workforce investment through a law establishing minimum wages for medical professionals, which raises the attractiveness of medical careers by ensuring systematic salary increases tied to education and specialisation. Portugal uses staffing benchmarks in program contracts and budget allocations, and Bulgaria and Greece use wage frameworks and incentive schemes to attract staff to underserved specialties or regions. Ireland invests in nursing and healthcare assistant posts under national frameworks, aligning budgets to staffing targets. Belgium embeds ratios into financing models for hospitals and long-term care institutions, while Austria relies on payer negotiations to steer investment. In Spain, staffing needs are factored into regional health budgets, even though national reports only provide guidance. Together, these mechanisms show how staffing levels can influence the scale and direction of financial commitments.

The training capacity link is visible in systems where accreditation and quotas depend on staffing requirements

Latvia expands training places for specialties with acute shortages or ageing workforces (Box 17), while Hungary aligns postgraduate quotas with staffing thresholds and service needs. Poland has expanded medical education by increasing admission limits, opening new medical schools, and raising the number of residency places to align training capacity with healthcare sector employment needs (Box 16). In Czechia, the national strategy explicitly increased medical training capacity following workforce analysis tied to staffing targets, while Portugal ties training places to institutional staffing and accreditation criteria set with the Medical Association. Ireland collaborates across ministries and higher education institutions to expand capacity in line with staffing forecasts, while Greece sets specialty training terms jointly with the Ministry of Education and professional bodies (Box 18). Belgium sets a quota for each community at the federal level defining how many physicians and dentists may enter specialty training within the national health insurance system. Communities then determine the number of medical and dental students admitted, applying a loss-rate formula for expected attrition, and may set sub-quotas to address specific needs and priorities. Austria balances federal and state priorities with professional input, ensuring training outputs reflect negotiated staffing needs. Romania integrates training and planning in its multi-annual Human Resources for Health (HRH) strategy, aligning educational expansion to masterplans. Here we see that staffing levels can be used not just as static requirements but as dynamic triggers for adjusting training pipelines.

In systems where staffing levels are not legally binding it is more difficult to establish a link between training, investment and planning

Spain relies heavily on regional autonomy: while national reports forecast professional needs, implementation of training and investment depends on each autonomous community (Box 19). Austria's lack of binding rules means planning and training are shaped by negotiations rather than enforceable standards, while Belgium faces challenges balancing federal quotas with regional implementation. Romania, though it has strong strategic documents, must contend with disparities between affluent and underserved regions, which limit the effectiveness of national plans. These examples show that without binding integration or systematic monitoring, the impact of staffing levels on planning, training and even investment can be fragmented or inconsistent.

In summary, these governance arrangements reveal a spectrum. Countries that have legally binding staffing levels such as Bulgaria, Hungary, Czechia, and Greece create enforceable or semi-enforceable links between staffing levels, workforce planning, investment flows, and training quotas. Finance-driven systems like Italy tie staffing to expenditure ceilings, making fiscal rules the central link between staffing and workforce development. Federal systems such as Belgium and Austria balance national quotas or frameworks with regional discretion, ensuring coordination but leaving room for variation. Decentralised systems like Spain rely on national reporting and strategic planning, with regions translating guidance into staffing, investment, and training action. Across all of the countries, however, Ministries of Health frequently work with Ministries of Education and professional bodies to expand or accredit training capacity, showing that staffing levels, whether binding or not, are a key driver of planning, budgeting, and education policies.

BOX 15 - Romania's health workforce approach connects planning and staffing

How it works

In Romania, health workforce planning and staffing levels are coordinated by the Ministry of Health, supported by the Inter-institutional Decision-Making Group. This body, which includes universities, the College of Physicians, professional organisations, and local authorities, aligns staffing levels with workforce planning using population, workload, and workforce profiling, clinical guidelines, and modelling approaches.

Link to planning and forecasting

The legal framework is extensive and anchored in Law no. 95/2006 on health reform, the Multiannual Strategy for the Development of Human Resources in Health 2022–2030, and the National Health Strategy 2023–2030. Thus, a strong link exists between forecasting and staffing decisions. County public health authorities collect monthly statistical data from health units and prepare community health reports, which are shared with the Ministry of Health and local partners. The National Health Insurance House, together with the Ministry, develops evaluation methodologies for healthcare institutions. This ensures that real-time data feeds into staffing and workforce planning, improving responsiveness.

Romania embeds workforce forecasting into its regional health service masterplans and the Multiannual HR Strategy 2022–2030. These strategies focus on:

- Generating health human resources (through training and recruitment).
- Managing and motivating the workforce.
- Strengthening governance of workforce planning.

This forward-looking approach links forecasting exercises directly to staffing requirements, aiming to reduce urban–rural disparities and adapt staffing levels to demographic and regional differences.

Summary

Romania provides an example where workforce planning is linked to staffing levels through national strategies, regional masterplans, and contractual mechanisms. Data-driven forecasting and inter-institutional coordination ensure staffing norms reflect workforce sustainability, while local flexibility allows adaptation to regional differences.

BOX 16 - How staffing levels are linked with investment decisions in Poland

How it works

Staffing levels are embedded in Poland's healthcare governance and financing frameworks, serving as key levers for workforce planning and resource allocation. The Ministry of Health is the central authority responsible for setting minimum employment standards for doctors, nurses, and midwives through legally binding regulations under the Law on Healthcare Services Financed from Public Sources. These staffing norms are not only a condition for healthcare providers to obtain contracts with the National Health Fund (NHF), the only public payer, but also serve as a mechanism to shape how resources are distributed across the system. By linking staffing requirements to reimbursement contracts, the state ensures that investment flows directly toward sustaining workforce adequacy.

Compliance

The enforcement of these standards aligns with system-level investment strategies, including the National Transformation Plan (NTP) and Provincial Transformation Plans (WPTs). The NTP provides the overarching framework, setting national objectives for modernising the health system and addressing workforce shortages, especially in specialties such as internal medicine, oncology, and general surgery where there is a high demand. The WPTs adapt these priorities to regional contexts, using health needs maps to identify gaps and directing investment toward local challenges. This dual system ensures that national workforce goals are balanced with territorial realities, guiding financial allocations toward recruitment, retention, and redistribution of staff.

Link to investment

Staffing levels also influence investment in training and professional development. For instance, minimum employment standards create demand for more healthcare professionals, which has led to a significant expansion in training opportunities. The number of residency places increased from 7,358 in 2015 to over 10,400 in 2023, reflecting the need to align educational capacity with staffing mandates. Similarly, systemic wage increases for healthcare workers were designed not only to improve retention but also to make medical careers more attractive, encouraging more students to pursue healthcare studies and remain in the profession. These measures demonstrate how staffing requirements directly trigger investment in workforce expansion, retention and educational infrastructure.

Summary

Poland demonstrates how legal staffing norms operate as investment levers by linking workforce adequacy to financing, service provision, and education. Meeting statutory staffing requirements is a precondition for contracting, while wage laws, targeted incentives, and expanded training capacity reinforce workforce sustainability and regional balance.

BOX 17 - From policy to practice: How Latvia links staffing levels and education

How it works

The Ministry of Health in Latvia is responsible for workforce planning and ensuring adequate personnel provision within the healthcare system. This planning is evidence-based, using population profiling, workload profiling, workforce profiling, clinical guidelines, and modelling approaches. Research organisations, professional associations, employers, higher education institutions, and other ministries are involved in developing recommendations for required personnel numbers and workloads. The results are integrated into policy documents and indicative targets or planning benchmarks, which set target values for staffing.

Latvia's Cabinet of Ministers regulations define the required availability of specific professionals (e.g. emergency teams, admissions staff), which medical institutions must ensure to qualify for state-funded contracts. These requirements are binding for medical institutions, and compliance is assessed by the National Health Service when contracts for state-funded services are concluded. Medical institutions are responsible for ensuring they meet these thresholds, while the Ministry of Health monitors implementation through mid-term and final assessments of policy indicators.

Link to education

A key feature of Latvia's approach is the direct link between staffing requirements and training capacities. The Ministry of Health determines the number of residency places and their distribution across medical institutions. This planning explicitly takes into account:

- Hospital needs for specialist training to meet Cabinet of Ministers' requirements for admissions staff.
- Shortage specialties, where additional residency places are allocated from the state budget.
- Aging professions (e.g., family doctors), where increased training intake is needed to replace retiring staff.
- This ensures that education capacity aligns with projected workforce gaps and regulatory obligations.

Nursing profession reforms

Latvia has also reformed its nursing profession to address shortages. The general care nurse role replaced narrow specialisations, enabling nurses to work across multiple settings and levels of care. In addition, the advanced practice nurse role was introduced, especially in primary care. To support this reform, the number of state-funded nursing education places was significantly increased, helping to address shortages and broaden workforce flexibility.

Summary

Latvia provides a clear example of how training capacities are strategically tied to staffing level requirements. By linking policy-driven minimum thresholds with residency planning and nursing reforms, Latvia ensures that workforce education responds to both immediate and long-term health system needs. This integrated approach balances regulatory enforceability with forward-looking workforce sustainability.

BOX 18 - How Greece links workforce planning and staffing levels

How it works

In Greece, workforce planning and staffing levels are coordinated by the Ministry of Health in collaboration with Regional Health Authorities (RHAs). The National Health System (NHS) provides universal access to primary, secondary, and tertiary care, and since 2019 has focused on aligning staffing needs with population health priorities, service demand, and regional disparities.

Workforce planning follows a structured methodology that assesses healthcare needs, maps the existing workforce, defines target staffing ratios by specialty and facility level, and validates plans regionally. Similar models are applied for nursing and allied professions, integrating factors such as attrition, training capacity, and projected retirements.

A major reform is the development of a Human Resources Information System (HRIS) in partnership with the World Health Organisation (WHO). The HRIS, expected to be operational by 2026, will enable real-time monitoring of workforce numbers, distribution, and skill mix to better link planning with day-to-day staffing management.

Link to planning and forecasting

Greece's system connects strategic forecasting with operational decision-making. RHAs collect data on staffing availability, service activity, and local needs, maintaining registers of auxiliary physicians to address urgent shortages. To improve recruitment and retention, particularly on the islands and in remote areas, the government introduced in 2024 a package of financial and non-financial incentives, including bonuses, housing support, relocation assistance, and accelerated career progression.

Forecasting also guides medical education and specialisation policies, with financial incentives for priority fields such as general practice, internal medicine, anaesthesiology, and emergency medicine. These measures aim to reduce specialty shortages and strengthen the sustainability of the health workforce.

Summary

Greece's workforce planning model directly links forecasting to staffing levels, combining data-driven planning, regional flexibility, and targeted incentives. The approach strengthens equity in service delivery and builds a foundation for long-term workforce sustainability through continuous monitoring and evidence-based decision-making.

BOX 19 - Linking staffing levels with training, investment, and workforce planning in Spain

How it works

Spain's governance of staffing levels is shaped by its decentralised health system, where the autonomous communities hold primary responsibility for planning, recruitment, and investment. Each region determines its workforce needs according to population characteristics and demand, while healthcare centres often develop their own staff planning models within regional regulations. At the national level, the Ministry of Health contributes by publishing periodic reports projecting the needs of doctors and nurses (e.g., *Oferta y necesidad de médicos especialistas en España 2023–2035* and *Necesidades Profesionales de Enfermería 2024*). These reports are not binding but provide a shared evidence base for workforce planning across the country.

Non-binding staffing framework

Because staffing levels are not legally binding at either the national or regional level, Spain's link between staffing, investment, and training is more flexible but also more fragmented. Regions include staffing-related investments in their annual budgets, and workforce expansion is financed at the community level. This allows each region to tailor investment decisions to local needs, such as addressing rural shortages or boosting nursing capacity. At the same time, the absence of mandatory national staffing thresholds makes it harder to ensure uniformity and equity across regions. Some communities may respond quickly to shortages by expanding hiring or investing in training capacity, while others do not.

The link to training and workforce planning

Training and workforce planning are connected through national and regional coordination mechanisms. The NHS Human Resources Commission brings together the Ministry of Health, autonomous communities, and other ministries (education, universities, defence, justice) to align training capacity with projected staffing needs. This collaboration helps determine the number of medical and nursing students admitted each year and ensures that training expansion supports long-term workforce planning. Yet, because the benchmarks are indicative rather than prescriptive, success depends on how actively each region engages with the recommendations.

Summary

Spain shows how non-binding staffing frameworks can still support planning, training, and investment if they are combined with strong national reporting and intergovernmental coordination. The flexibility allows regions to adapt quickly to local realities, which is an advantage in a decentralised system. However, the lack of legal enforceability also creates challenges in achieving consistent workforce distribution and investment across the country, making it more difficult to guarantee equitable access to care nationwide.

Are staffing levels legally binding?

In several countries, staffing levels are legally binding and anchored in legislation or regulatory decrees. Out of the nine countries that define staffing levels at the system level, five have staffing levels that are legally binding (Bulgaria, Hungary, Poland, Greece and Czechia). In Bulgaria, staffing levels are set by regulatory requirements and legally binding contracts between the National Health Insurance Fund (NHIF) and professional organisations, with the NHIF and the Executive Agency for Medical Supervision enforcing compliance (Box 14). Hungary enforces minimum staffing requirements through legislation such as the 2003 Decree on healthcare services, making compliance obligatory at licensing and during inspections. Poland mandates binding staffing norms through Ministry of Health ordinances and additional executive regulations in hospitals, particularly for nurses, linking them directly to provider contracts with the National Health Fund (Box 16). Greece also embeds staffing requirements in law and presidential decrees, specifying hospital staffing by specialty and monitoring recruitment through centrally approved posts. In Czechia, the overarching national staffing target serves as a non-binding benchmark reviewed biennially; however, the specific staffing ratios in the health directive are legally binding and enforced. These binding rules establish a floor for safe care and ensure sufficient staffing is a condition for service provision.

Compliance mechanisms vary but are typically monitored by state authorities, payers, or supervisory agencies

In Hungary, authorities verify adherence to staffing thresholds during licensing and operational inspections, checking both qualifications and numbers (Box 20). Poland's National Health Fund monitors compliance as part of its contracting process, ensuring providers meet workforce requirements for reimbursement. In Greece, oversight is shared between the Ministry of Health and Regional Health Authorities, supported by digital reporting systems such as the BI-Health platform. Bulgaria combines legal obligations with social partnership agreements and compliance is monitored by both the National Health Insurance Fund and the Executive Agency for Medical Supervision (Box 14). These systems highlight the use of strong regulatory tools backed by contractual enforcement or inspections to secure compliance.

Fiscal oversight plays a critical role when staffing is linked to expenditure

Italy enforces staffing levels indirectly through strict expenditure caps, monitored jointly by the Ministry of Health and the Ministry of Finance. Compliance is checked via the national technical committee (*Tavolo tecnico per gli adempimenti*), which reviews whether regions respect budgetary ceilings tied to essential care delivery. While this does not prescribe ratios at the clinical level, it legally constrains workforce growth and makes staffing subject to financial accountability (Box 13). Similarly, in Bulgaria, staffing levels are embedded in contractual agreements tied to financing negotiations, making compliance enforceable via funding channels. These two examples demonstrate how legal binding can be embedded not only in ratios but also in financial ceilings and reimbursement structures.

Regional or institutional actors take on compliance responsibilities, even where central benchmarks exist

Latvia requires medical institutions to meet Cabinet-defined minimums for emergency teams and admissions departments, with the National Health Service checking compliance during contracting. However, outside of these areas, indicators are treated more as guidelines (Box 17). In Romania, county public health authorities and the National Health Insurance House oversee data collection and evaluation, ensuring providers meet criteria set jointly with the Ministry of Health. In Ireland, compliance with staffing ceilings is monitored monthly, but responsibility for allocation rests with local HR and planning officers within the HSE (Box 29).. These models blend binding and flexible enforcement, allowing some autonomy while keeping central checks in place.

Non-legally binding solutions that still influence staffing decisions

Spain's autonomous communities set their own workforce needs, with national reports providing guidance but not enforceable standards (Box 19). Portugal relies on indicative staffing benchmarks and staffing maps used for planning, contracting, and internal audits, but without legal penalties for non-compliance (Box 21). Ireland's modelling and projections also serve as strategic tools rather than binding rules. These examples show that even where staffing solutions are not legally enforceable, guidelines, reference values, and strategic targets can play an important role in shaping planning, budgeting, and workforce capacity over time. Austria's frameworks are set by federal and state governments and sickness funds and grounded in evidence (population/workload/local HWF profiling, guidelines, scientific literature). As Austria does not specify statutory minimums or a national compliance regime, it is closer to the non-binding, payer/contract-led end of the spectrum.

Across Europe, staffing rules are a mix of legally binding and non-binding approaches, reflecting different governance traditions. Binding models, such as in Hungary, Poland, and Greece, enforce minimum ratios through law, inspections, or payer contracts, while countries like Spain, Portugal, and Austria rely on guidelines, benchmarks, or negotiated frameworks that steer practice without legal sanctions. This blend shows how systems balance enforceability with flexibility, adapting regulation to national contexts and health system needs.

BOX 20 - How Hungary embeds staffing levels in law and policy

How it works

Staffing levels in Hungary are anchored in legislation and ministerial decrees, making them legally binding across the health system. The Health Ministry determines the minimum professional conditions for the provision of services, including the number and qualifications of staff, under the authority of the Health Act (Act CLIV of 1997). These standards are detailed in Decree No. 60/2003, which defines the minimum staffing requirements for healthcare institutions. Compliance is mandatory as healthcare providers must meet these thresholds both to obtain operating licenses and to continue functioning legally. This binding approach reflects the view that safe patient care cannot be delivered below defined staffing levels.

Compliance

The legally binding character is reinforced through strict monitoring and enforcement mechanisms. Authorities assess compliance during licensing procedures and through regular or ad hoc inspections, reviewing both staff numbers and qualifications. Compliance checks

include whether staff are registered professionals and actively participating in mandatory training. This oversight ensures that staffing rules are not mere formalities, but active requirements embedded into daily operations. Providers that fail to meet thresholds risk penalties or loss of licenses.

These binding minimums also shape workforce planning and investment decisions. To maintain compliance, providers must secure sufficient qualified staff, which drives demand for training and recruitment. At the national level, minimum thresholds are factored into state-funded specialist training quotas and nurse training needs assessments. Hungary's budget law directly supports these obligations through allocations for payroll, scholarships, and grants, while the National Health Insurance Fund (NEAK) incorporates personnel costs into provider financing. In effect, staffing laws create an investment obligation, linking funding flows to workforce adequacy.

Training policy is also influenced by these legally binding standards. The Resident Support Program (Government Decree 162/2015) expands postgraduate training slots to ensure that retiring staff are able to be replaced and to meet the demand created by the minimum thresholds. For undergraduate education, admission quotas are more flexible, but postgraduate programs are tightly tied to workforce planning needs. This ensures alignment between staffing laws, training capacity, and long-term human resource sustainability.

Summary

The case of Hungary depicts how legal enforceability transforms staffing levels into a structural pillar of health governance. By embedding thresholds into law, monitoring compliance rigorously, and linking them to financing and training policies, Hungary ensures that workforce adequacy is non-negotiable. This approach provides stability and accountability but requires sustained public investment and careful balancing of flexibility to respond to changing healthcare needs.

BOX 21 - The strategic influence of non-binding staffing solutions in Portugal

How it works

Staffing levels in Portugal are not legally binding, yet they play a strategic and influential role in shaping health workforce planning, investment, and training. The Central Administration of the Health System (ACSS) provides national-level guidance by monitoring, recruiting, and planning human resources, while the Executive Board of the NHS adapts these guidelines to regional contexts. Professional associations, such as the Portuguese Medical Association and the Nurses' Association, issue technical and ethical recommendations, which, although non-binding, are widely used as reference points for safe and effective staffing. These advisory benchmarks influence how hospitals and local health units organise care delivery, allocate staff, and plan future recruitment.

Staffing maps

One key tool is the development of staffing maps by NHS institutions. The maps serve as planning instruments that help to define the number of healthcare professionals needed to provide services in line with an institution's profile and activity. They directly inform budgetary allocations and program contracts, linking workforce expectations to funding flows. Similarly, documents like the Hospital Referral Network for Cardiology translate workload profiling into reference values for required full-time equivalents (FTEs), guiding service provision without imposing strict mandates.

Non-binding staffing framework

Non-binding staffing levels also shape training and professional development. For example, in the medical internship process, the ACSS and Portuguese Medical Association assess training capacity against staffing needs, which determines the number of residency slots by specialty and institution. Accreditation of training institutions depends partly on available staff, while residency-to-supervisor ratios are guided by professional standards. These processes ensure that workforce planning, and educational capacity remain closely aligned, even without binding quotas.

Investment decisions are similarly influenced through strategic frameworks and incentives. The Health Human Resources Plan 2030 translates reference staffing benchmarks into long-term strategies for recruitment and retention. Government measures, such as financial incentives for underserved areas or full-time dedication bonuses, are designed in response to staffing reference values and workforce projections. While compliance is monitored through administrative reviews, clinical audits, and professional councils' oversight, non-compliance has no legal consequences, making the approach flexible yet effective in steering investment priorities.

Summary

Portugal provides an example as to how non-binding staffing solutions can still shape system behaviour. By embedding staffing benchmarks into planning instruments, investment frameworks, and training accreditation processes, Portugal ensures that workforce supply remains responsive to population needs, although the supply-demand balance still shows some limitations, particularly in GP/primary care. The lack of legal enforcement offers flexibility, while the widespread reliance on reference values creates a de facto standard that influences staffing decisions across the health system.

Sectoral level: hospitals, primary care and other sectors

What are the governance solutions for staffing levels, how are they agreed upon and who is involved?

In the hospital sector, governance solutions are typically anchored in legal frameworks, planning tools, and negotiated agreements. In Slovakia, Greece and Hungary, Ministries of Health establish legally binding minimum thresholds, often tied to licensing and compliance inspections. Bulgaria, Belgium, and Luxembourg rely on a mix of legislation and negotiated frameworks with insurance funds or professional bodies, where compliance is monitored and sanctions or loss of funding may follow non-compliance (Box 22). Austria and Germany integrate staffing rules into broader hospital legislation and financing frameworks, including DRG-based payment systems, thereby making noncompliance with staffing requirements a condition for a reduced reimbursement (Box 23). While this approach constrains the flexibility originally embedded in DRGs, it reflects a deliberate policy choice to balance efficiency incentives with safeguards for quality of care, patient safety, and workforce sustainability. These arrangements are agreed at national level by central ministries, health insurers, and social partners, ensuring consistency across the system.

In primary care, governance is generally more decentralised and flexible. Slovakia and Spain use modelling and regional evaluations of provider networks, with regional authorities deciding whether to follow national recommendations (Box 24). Portugal sets staffing levels through strategic planning bodies like the ACSS, but these are mainly guidelines rather than binding rules (Box 12). Austria embeds primary care staffing in structural plans or quotas negotiated between health insurance funds, governments, and professional associations. In Bulgaria, staffing levels are legally binding, tied to organograms and national frameworks, with ministries and regional health authorities overseeing allocation. Agreement in primary care often involves negotiations between ministries of health, education, insurers, and professional chambers to balance supply, demand, and access.

In other sectors such as ambulatory, specialist, nursing, and dental care, governance mixes central rules with regional flexibility. Slovakia applies fixed but adaptable formulas for ambulatory care, while Austria sets national frameworks that regions or providers cannot undercut but may supplement. Spain uses reference values, modelling, and performance contracts rather than enforceable thresholds, leaving room for regional adaptation for specialists. Across these non-hospital sectors, solutions are agreed through joint commissions, advisory bodies, and regional health authorities, involving governments, insurers, professional bodies, and providers, ensuring staffing rules reflect both system-level goals and local realities.

Norway highlights how governance differs in a Nordic model where trust-based, decentralised planning plays a bigger role. In hospitals, Norway sets staffing expectations mainly through national legislation, collective agreements, and professional standards, but implementation is delegated to regional health authorities and individual hospitals, which adapt to local needs. In primary care, municipalities have responsibility for staffing, guided by national frameworks and financial transfers, with professional associations (e.g., the Norwegian Nurses Organisation, the Medical Association) involved in shaping guidelines and good practice norms. Agreements are therefore less about hard thresholds and more about negotiated standards, capacity planning, and ensuring that investments and training keep pace with demographic and care needs.

Compared to system-level governance, which typically sets overarching rules, expenditure ceilings, or national workforce strategies, sectoral governance is more fragmented and operational. At the system level, countries like Italy or Greece set nationwide caps on personnel or posts through laws and decrees, often driven by ministries of health and finance with limited room for local adjustment. In contrast, sectoral governance in hospitals and primary care introduces more detailed tools. This includes, for example, workload profiling in Slovakia, referral networks in Portugal, or DRG-linked rules in Austria. Each of which directly shape staffing in specific settings. Agreement processes also differ while system governance is highly centralised and political, sectoral governance involves broader coalitions including hospital associations, insurers, professional chambers, and regional authorities. This means sectoral solutions are often more practical and context-sensitive, but also more varied, reflecting the very distinct service delivery challenges of hospitals and primary care.

BOX 22 - Governance of staffing levels in Belgian hospitals

How it works

The governance of hospital staffing levels in Belgium is shared between the federal government and the regions. The federal government sets the minimum legal requirements for staffing through Royal Decrees, which define fixed ratios for wards, units, and care programs. These ratios act as accreditation standards: if a hospital fails to comply, it cannot receive authorisation to operate that service and loses the associated funding. The regions are responsible for inspection and enforcement. They verify compliance with the federal standards and, in theory, may revoke a hospital's authorisation if ratios are not met. Moreover, regions can set stricter requirements beyond the federal minimums, creating room for variation across the country.

Multi-level stakeholders

The governance process incorporates the perspectives of multiple actors. The hospital sector is formally consulted through an advisory committee, ensuring that providers' views are taken into account. In some cases, the regions are also included in discussions when developing or adjusting standards, reflecting Belgium's federal structure. This shared approach aims to balance uniform national requirements with the ability to address regional and local needs.

Legal minimum

Although the standards are designed as minimal thresholds, Belgian hospitals typically employ more staff than the legal minimum. One of the reasons lies in the financing arrangements: the federal government not only sets the rules concerning the legal minimum staffing but also funds hospitals.

To supplement the minimum staffing levels, the federal budget subsidises more staff than required by the Royal Decrees. In addition, hospitals can voluntarily choose to recruit more personnel in some services. This creates a gap between the legal framework and the reality: while compliance with the minimum is mandatory, the actual workforce composition usually exceeds it.

Hospitals that fail to uphold the staffing ratios theoretically risk losing authorisation and funding. In practice, enforcement varies depending on regional policy priorities. Nonetheless, the binding nature of the Royal Decrees and the role of the regions in the inspection procedures ensure that staffing standards remain central to hospital governance.

Summary

Belgium's governance of hospital staffing levels combines federal standard-setting, regional enforcement, and hospital sector consultation. This layered governance structure reflects Belgium's federal complexity while safeguarding patient safety and service quality.

BOX 23 - Germany's governance model for hospital staffing levels

How it works

Governance solutions for hospital staffing levels in Germany are structured around a dual regulatory approach, combining federal legislation with delegated decision-making. For somatic hospitals, staffing levels are set directly through a regulation issued by the Federal Ministry of Health (BMG), which establishes binding minimum staffing requirements across key hospital units. These thresholds are based on empirical workload research, targeting wards with particularly high patient demand. For psychiatric units, responsibility is delegated to the Federal Joint Committee (G-BA), the country's highest self-governing body in healthcare, which uses acuity-based assessments to determine staffing needs. This division ensures that both somatic and psychiatric care are governed through evidence-based rules adapted to their distinct service realities.

Multiple stakeholders

The process of agreeing on staffing levels involves multiple actors. The BMG sets binding requirements for somatic care after consultation with research experts and professional stakeholders, while the G-BA develops psychiatric standards through a structured negotiation process. The G-BA is composed of representatives from statutory health insurers, hospitals, and physicians' associations, with patient representatives holding advisory roles. This inclusive structure ensures that staffing regulations reflect not only federal policy priorities but also professional standards, payer perspectives, and patient needs. Research institutions provide the data underpinning staffing norms, while professional associations ensure alignment with clinical practice.

Compliance

Compliance with staffing rules is enforced through reporting and sanctions. Hospitals must report their staffing levels and the number of understaffed shifts quarterly and annually. If monthly averages fall below mandated thresholds, financial sanctions are applied under agreements linked to the German Diagnosis-Related Groups (G-DRG) system. These rules are non-negotiable at the local level—hospitals may exceed minimums but cannot go below them, except under extraordinary circumstances (i.e. COVID-19 pandemic). This centralised, legally binding model secures baseline staffing, while still allowing hospitals to invest in higher staffing ratios if desired.

Summary

Germany's governance of hospitals combines federal authority, corporatist negotiation, and strict enforcement, making staffing regulations a cornerstone of hospital workforce management.

BOX 24 - The governance of staffing levels in Slovakian primary care

How it works

Staffing levels in primary care are governed centrally by the Slovakian Ministry of Health, which applies a doctor-to-patient ratio as the baseline indicator. Using population profiling, workload profiling, health workforce profiling, and modelling approaches, the Ministry evaluates whether the expected number of patients in each region can be adequately served by the available number of general practitioners and primary paediatricians. This process is formalised through the evaluation of the optimal network of providers, an annual exercise introduced in 2022. The evaluation identifies unmet demand and highlights districts at risk of poor access, serving as an early warning system for workforce shortages.

Multi-level governance

This form of governance relies on both national oversight and regional involvement. While the Ministry of Health develops the model and runs the annual evaluation, regional authorities have the discretion to agree or disagree with the outputs. This means that while the Ministry provides the technical evidence base, final decisions on whether to accept the evaluation results rest with the regions. Importantly, healthcare providers themselves cannot adjust staffing levels, as this remains under national jurisdiction. The approach ensures national consistency but allows regional authorities to incorporate contextual realities, such as geographic distribution or local workforce mobility.

Compliance

Staffing governance is also linked to incentives for underserved areas. When evaluations identify districts with unmet demand, the Ministry of Health allocates non-repayable financial contributions to support the establishment of new clinics, which can be applied for by GPs, paediatricians, or existing providers. This mechanism helps target investment where it is most needed, while also encouraging professionals to relocate or expand services in underserved regions. Future plans include strengthening the competencies of general practitioners and reducing administrative burdens to improve retention.

Summary

Although staffing levels are not legally binding in primary care, the Slovak model combines central monitoring, regional negotiation, and targeted financial incentives to maintain equitable access to primary care.

Is there a link between hospital staffing levels and HWF planning, investment and training capacities?

Many countries demonstrate a clear link between hospital staffing levels and workforce planning, though the strength of this connection varies

In Slovakia, Czechia, and Bulgaria, staffing ratios and workload-based models for hospitals are explicitly used to forecast future workforce needs, with ministries and professional bodies adjusting quotas and identifying shortages in specific regions or specialties. Greece and Germany (Box 23) embed staffing levels for hospitals directly into system planning, where national thresholds influence the allocation of posts and budgetary envelopes, ensuring that planning aligns with legally binding requirements. By contrast, countries like Spain, Portugal, and Ireland rely on modelling and indicative staffing benchmarks, which are integrated into broader workforce strategies rather than enforced at the sectoral level (Box 12).

Staffing levels also shape investment decisions across sectors

In Belgium (Box 22) and Austria (Box 9), compliance with minimum ratios is tied to hospital financing and authorisation, creating a direct incentive for investment in personnel. Luxembourg links staffing agreements to hospital budgets negotiated with insurers, ensuring funding reflects both service demand and staff workload (Box 10). In Portugal and Ireland, although staffing levels are less rigid, they guide annual performance contracts and budgeting frameworks, helping target resources towards areas of greatest need. Even in more decentralised systems like Sweden and Spain, national frameworks and regional planning use staffing benchmarks to prioritise investment in underserved regions or specific types of care.

The link with training capacities is most explicit where staffing thresholds highlight shortages

Bulgaria and Greece adjust university admission numbers and specialisation places in medicine and nursing to meet gaps identified through staffing levels in hospitals (Box 18). Austria and Slovakia are developing forecasting systems to connect hospital staffing requirements with education pipelines, while Ireland and Portugal expand training places in collaboration with higher education institutions in line with projected demand. Norway and Spain rely on municipalities and regions to feedback local shortages into national workforce education planning, ensuring training supply reflects service delivery needs.

Across all these cases, staffing levels act as both a planning signal and a lever for aligning investment and training policies with sectoral service demands.

Are staffing level solutions in primary care an emerging and promising trend?

Defined staffing level solutions in primary care support the shift towards more community-based care

In Portugal modelling tools and patient-to-doctor ratios help identify underserved areas and channel investment or incentives to expand coverage (Box 12), while in Bulgaria, legally binding frameworks tie staffing levels to organograms and training quotas, ensuring posts are filled across regions. Austria integrates staffing plans into broader structural frameworks with insurers and professional associations, gradually enabling role expansion for general practitioners, nurses, and allied professionals. Countries like Spain and Norway demonstrate how flexible, regionally adapted planning can reinforce integrated community care by linking staffing in primary care to local demand, while also opening space for team-based and task-sharing approaches.

Staffing level solutions in primary care seem to be emerging as a promising trend across the EU and Norway. These solutions can not only strengthen equity of access but also accelerate the expansion of roles in primary care of nurses, family doctors, and multidisciplinary teams. This in turn allows for the delivery of more preventive, continuous, and population-oriented care outside hospitals.

Chapter 5: What do we know about the evaluation of staffing levels?

System level

Do countries see opportunities and risks of setting staffing levels at the system level?

Across all surveyed countries, regardless of whether they define staffing levels or not, most see opportunities in defining staffing levels, mainly to improve safety, quality, and equity, and they are building the data and methods to do it more efficiently (See Table 2). Slovenia, for example, plans to introduce a unified, workload-based methodology and develop national standards in 2026 (Box 26). Slovakia is updating rules to cap patients per nurse/doctor and is mapping regional minimum/optimal thresholds in ambulatory care. Spain is piloting a patient-acuity classification, 'Safe Nurse Staffing', with EU technical support to align nursing numbers with care needs. It intends to apply the Safe Nurse Staffing classification across hospital, primary, emergency (and long-term) care. Hungary is designing a dual framework that combines workload distribution with operational minimums for both inpatient and outpatient services. Similar planning tools and strategies are advancing in Portugal through the Joint Action HEROES and their 2030 HRH plan, in Estonia where they are in the process of building a HWF forecasting model for primary care (Box 25), in Latvia where they are creating a 2025–2029 workforce strategy and a unified register, and in Malta and Greece through a planning tool and a single data repository. Romania is developing a multi-year HRH strategy and working on capacity building, Poland is mapping health needs and expanding education pathways, Czechia is reviewing its staffing directive, and Bulgaria is developing a national monitoring methodology.

The risks most often cited in relation to staffing levels include their rigidity, weak data, and misalignment with real-world workload or geography. Norway, for example, explicitly argues that rigid staffing levels can work against flexibility, hence its emphasis on task-shifting (Box 6), guideline updates and cross-trained education pathways. Sweden's focus on a National Health Competence Council and a primary-care-led transformation also emphasises skills and organisation over fixed numbers (Box 8). Slovakia cited that its existing rulebook (an older ministerial regulation specifying how staffing levels for doctors and nurses should be calculated) is outdated while Slovenia's Blue Books of Staffing Levels for doctors and nurses exist but were not adopted because the original methodology could not be verified (Box 26). Latvia still needs to operationalise its model inside ministry systems and Hungary is updating minimum requirements to avoid having to rely on obsolete headcounts. Poland's heavy education and migration levers are helpful but require constant calibration and Greece highlights persistent shortages in remote areas even with the incentives it offers. Taken together, it seems that countries value staffing levels for safety and transparency but tend to caution that poorly measured or overly rigid rules can freeze inefficiencies and worsen shortages.

In general, there appears to be a momentum toward defining staffing levels or equivalent guardrails, but the direction seems to be more data-driven and conditional rather than purely ratio-based. The emerging consensus appears to lean towards anchoring staffing in measurable workload, patient acuity, and risk adjustment. All of which is supported by forecasting models, unified registers, and better governance, while preserving flexibility for task-shifting and new care models. Countries give the impression of seeing the opportunity to raise quality and transparency and to target shortages more precisely, but they also appear to be aware of the risk that strict mandates could reduce efficiency or hinder innovation. Therefore, some countries are pairing the minima they do set with adaptive tools, regional tailoring, and ongoing evaluation.

BOX 25 - Estonia's path toward possible staffing levels

Where they are going

Estonia is actively exploring future workforce planning solutions. The Qualifications Board published a workforce needs analysis in late 2024, providing evidence on long-term health workforce gaps. In parallel, the Ministry of Social Affairs is developing a health workforce forecasting model under the Joint Action HEROES, with the aim of covering the primary care sector by the end of 2025. This marks an important step toward more structured and data-driven planning.

What have they done

Recent reforms in the nursing profession illustrate Estonia's incremental approach. In 2025, five separate nursing legislative acts were consolidated into one, simplifying regulation and clarifying nurses' scope of practice. According to the Estonian Nurses Union, the updated framework strengthens nurses' rights to work independently, improving efficiency and task-sharing. These changes suggest Estonia is prioritising flexibility and role expansion rather than fixed numerical ratios.

The National Health Plan 2020–2030 highlights workforce development as a priority but does not commit to introducing staffing levels. Instead, the government has announced plans to prepare a specific action plan within the next year to address the workforce crisis. This plan is expected to draw on the forecasting model and sectoral evidence to identify tailored measures.

Summary

Estonia sees potential in strengthening workforce planning through forecasting, role development, and targeted action plans, but has not yet committed to mandatory staffing levels. Future decisions will likely balance the benefits of clearer minimum standards with the risks of reducing flexibility in a small health system.

BOX 26 - Workload data as the basis for future staffing in Slovenia

Where they are

Slovenia does not currently apply binding minimum staffing thresholds. Staffing is instead shaped indirectly through negotiations between the Health Insurance Institute of Slovenia (ZZZS) and professional organisations, which define covered service programs, required capacities, and available funding. While this process aligns resources with population needs, it does not translate into enforceable norms for staff-to-patient ratios.

Where they want to go

In May 2025, an amendment to the Health Services Act (ZZDej-N) was adopted, which introduces the obligation to measure effective workload in all public healthcare institutions. Based on this, each institution will monitor the workload of its employees and collect data that will allow for more accurate determination of the necessary staff to ensure the delivery of high-quality healthcare services.

The Ministry of Health is currently in the process of drafting regulations to define a uniform methodology for measuring the daily workload of healthcare workers or healthcare assistants who provide healthcare services in public healthcare institutions at the primary, secondary, and tertiary levels of healthcare, the preparation of workload analyses, and the method of reporting.

Although professional associations have already developed indicative benchmarks, i.e. as the Blue Book of Staffing Levels for Physicians and Dentists and the Staffing Levels in Nursing and Midwifery, these have not been formally adopted due to methodological gaps. The new workload-based approach aims to close this gap by producing verifiable, comparable data that can underpin legally binding staffing levels. By 2026, Slovenia plans to use these data to define national staffing standards that ensure safe, high-quality, and cost-effective care, marking a shift from negotiated budgets to evidence-driven workforce governance.

Summary

Slovenia currently avoids staffing thresholds, relying instead on negotiated service programs and funding allocations. However, ongoing reforms to measure effective workload signal a shift toward data-driven workforce planning, which may pave the way for introducing staffing standards in the future.

Table 2 Opportunities and future staffing level plans

Country	Future plans	Predicted outcomes / implications
Belgium	Ongoing quota planning for physicians/dentists; future scenarios updated regularly.	Reduced risk of shortages/imbalances; quotas act as indirect staffing guardrails.
Bulgaria	Create national methodology for monitoring/evaluating health personnel (RRP).	Standardised, regulated assessments; groundwork for sector-level staffing definitions.
Czechia	Evaluate Health 2030 in 2025 (extend to 2035); consider reform of staffing directive.	Possible modernisation of staffing rules; longer-term strategic alignment.
Germany	—	—
Estonia	Build HWF forecasting model (primary care by end-2025); design an action plan to address workforce crisis.	Earlier detection of gaps; more coordinated primary-care staffing; foundation for broader sector planning.
Ireland	Annual “pay and numbers” process reviewed monthly/year-end.	Budget-aligned staffing; responsiveness depends on fiscal space and linkage to service need.
Greece	Developing a national HR Information System (HRIS) to monitor numbers and skill mix; incentives for underserved areas and critical specialties.	More accurate matching of staff to demand; improved distribution and retention; foundation for evidence-based minima where appropriate.
Spain	National HRH Strategy; patient-acuity classification (Safe Nurse Staffing) with EU TSI support; apply across hospital, primary, emergency and long-term care.	Better match of nursing resources to patient need; consistent national approach; supports quality and safety improvements.
Italy	Experimental national methodology (DM 24.1.2023) to set min-max staffing standards in hospital/home care (2022–2024) in 9 regions.	Evidence base for broader roll-out; gradual movement toward standardized minimums in selected services.
Latvia	Implement 2025–2029 Health Workforce Strategy; unified register; reforms to tariffs and work environment (e.g., 3-shift nursing).	Higher-quality data and planning; tariffs incorporate required HR; better retention and safer staffing patterns.
Luxembourg	—	—
Hungary	Dual framework combining workload distribution (WLD) and operational minimums (OSL) across inpatient/outpatient services; update minimum headcounts.	Adaptive, workload-based staffing with protected minimums; supports reallocation and medium-term stability.

Malta	Implement 2022–2030 Health Workforce Strategy; national single data repository; planning tool in use.	Stronger forecasting and allocation; better, faster decisions on minimum safe staffing where needed.
Austria	Considering adjustments to minimum personnel thresholds for certain specialised hospital units; introduction of systematic monitoring of hospital staffing levels.	
Poland	Ongoing maps of health needs; expansion/quality upgrades in medical education; planned update to education standards and simulation capacity.	Increased and better-prepared supply; improved regional planning; indirect support for meeting minimum staffing needs.
Portugal	2030 HRH Plan; risk stratification; Local Health Units and Integrated Responsibility Centres; full-time dedication regime.	More precise shortage/excess estimates; flexible cross-unit staffing; incentives and governance that support safe minimum cover.
Romania	Tools for workforce and service planning; governance upgrades via NRRP (training, HR policies, standards).	Stronger HR management capacity; clearer staffing standards and improved retention pathways.
Slovenia	In the process of drafting regulations to establish a uniform methodology for measuring daily workloads.	Data-driven, transparent staffing norms; closer alignment of staff to service demand; improved efficiency and quality.
Slovakia	New regulation to set max patients per nurse/doctor and other roles; develop regional networks with minimum/optimal thresholds in ambulatory care.	Updated ratios that reflect current practice; clearer regional need estimates; more balanced distribution of staff.
Finland	Ongoing planning for continuity of treatment in primary care (may involve staffing discussions).	Potential refinement of staffing requirements tied to access/continuity goals.
Sweden	System transformation with primary care as hub; National Health Competence Council coordinates skill supply.	Improved coordination and continuity; emphasis on competence supply over fixed ratios.
Norway	Use forecasts for overall needs; prioritize task-shifting and flexible organization rather than fixed staffing levels.	Maintains flexibility; supports role substitution; unlikely to adopt rigid sector-wide minima.

Do countries evaluate their staffing levels at the system level?

Health workforce planning can effectively serve as a bridge for evaluating staffing levels in healthcare by providing a structured, evidence-based framework for assessing whether the number, mix, and distribution of personnel align with population health needs and service demand. Several countries highlight the role of the Joint Action HEROES in improving models and data and the potential of this work to inform staffing decisions. Through the use of forecasting models, workload analysis, and skill-mix assessment, workforce planning enables organisations to move beyond simple headcounts and systematically determine staffing adequacy, shortages, or surpluses. It also links staffing evaluation to broader health system objectives - including quality of care, patient safety, access, and financial sustainability - ensuring that staffing decisions support strategic priorities rather than isolated operational concerns. Moreover, because workforce planning is iterative, it supports continuous monitoring and adjustment, allowing healthcare systems to respond to shifts in demographics, disease patterns, and service utilisation. As a result, health workforce planning functions not only as a tool for long-term policy formulation but also as a practical mechanism for informing day-to-day staffing decisions, thereby bridging the gap between system-level planning and organisational workforce deployment.

Many countries that define staffing levels have developed formal mechanisms to evaluate staffing levels by combining data collection, forecasting, and long-term workforce strategies. Poland, for example, has a legally mandated “map of health needs” that is updated annually with demographic, epidemiological, and workforce analyses (Box 27). This feeds directly into both national and provincial transformation plans, ensuring workforce planning is embedded in broader health system reforms. Hungary has also established strong structures, with the National Hospital Directorate General and the National Centre for Public Health and Pharmacy jointly producing reports and forecasts on trends such as retirement, migration, and supply gaps. Romania uses a relatively institutionalised and data-driven approach that links staffing evaluation to national strategies and regional planning (Box 28). Spain, although it does not define staffing levels at the system level, also integrates workforce planning into a national strategy aligned with the WHO-Europe framework, while also experimenting with patient acuity-based staffing models for nurses. Other countries, such as Ireland and Greece, rely on indicators, tools, strategies to align workforce supply with population needs, highlighting a clear commitment to defining and evaluating staffing levels at the system scale.

Several countries adopt a lighter or more experimental approach to evaluation. Bulgaria, for example, has opted for a less formal, expert-driven model of evaluation without a standardised methodology (Box 31). Latvia has piloted a forecasting and mapping model with EU support, but its routine integration into health planning is still pending. Slovenia has produced the Blue Book of Staffing Levels for doctors and nurses, which sets out minimum requirements, but these standards remain unofficial because the methodology has not been fully validated (Box 26). Italy has only recently introduced an experimental methodology through the 2023 Ministerial Decree that has been applied in several regions but remains limited to hospital and home care. Estonia is in the process of developing a forecasting model that will cover primary care by 2025 (Box 25), while Sweden has created a National Health Competence Council to coordinate planning across regions and institutions. Portugal does not currently have a strategy for assessing staffing levels but does have a few mechanisms in place that allow for some level of assessment. These include regular national workforce statistics compiled by health authorities, workforce data in budget and planning documents, internal organisational plans of health units, and regional needs assessments using indicators like provider-to-population ratios, waiting times, and coverage levels to allow a partial and indirect assessment of staffing adequacy. These cases show countries building capacity for system-level evaluation, although with varying degrees of formalisation and national coverage.

Czechia also fits into this category of countries with evolving but not yet fully institutionalised evaluation mechanisms. Staffing level legislation in Czechia is updated roughly every two years, mainly to reflect broader reforms, new service types, or changes in professional education rather than to revisit underlying staffing methodologies. Recent adjustments include new requirements for integrated long-term care and updates following changes in medical specialties. However, these revisions do not amount to a systematic evaluation of the core formulas and ratios behind staffing norms. A comprehensive review is planned for 2026, informed by upcoming HSPA EG work, though its scope will depend on the priorities of the new government.

Across countries, some mechanisms for evaluating staffing levels do exist, but few are able to link these assessments directly to demonstrable improvements in health system performance. This may reflect the varying levels of advancement and implementation of staffing solutions across health systems, as well as differences in data availability and evaluation capacity. At present, only Ireland (Box 29) and Germany (Box 30) have developed comprehensive, outcome-oriented evaluation frameworks capable of generating this level of insight. The value of these cases is not in the specific figures they produce, which cannot be transferred across systems, but in showing how countries can design robust evaluation methods that connect staffing policies to their real effects, thereby strengthen future decision-making.

BOX 27 - A legally anchored system-level evaluation approach in Poland

How they evaluate

Poland evaluates health workforce staffing needs through a statutory Map of Health Needs led by the Ministry of Health under the 2004 Act on health services financed from public funds. The map is refreshed annually and combines demographic and epidemiological analysis with a stock-and-flow view of resources, i.e. facilities, services and the healthcare workforce, to identify national and provincial priorities. Findings are then entered into a National Transformation Plan, while each province (voivodeship) prepares a matching provincial transformation plan. Voivodes (regional governors), together with the NFZ (National Health Fund) branches and founders of medical entities, also conduct ongoing situational assessments to address emerging gaps.

The annual cycle draws on the latest available datasets and emphasises use and performance, not only headcounts. It reviews utilisation of services, capacity, and observed bottlenecks and then recommends actions by geography and care area. This creates a consistent, repeatable feedback loop between evidence (maps), planning (national/provincial plans), and operational adjustments (regional stakeholder reviews). Because the instrument is embedded in law, updates to priorities and actions are routinised rather than ad-hoc.

Evaluation linked directly to education and training

Poland ties its system evaluation directly to supply pipelines. Each year the Ministry of Health sets admission limits for medical and dental faculties and allocates residency (specialty) positions, considering university capacity (staffing, equipment, premises) and mapped population needs. Recent reforms strengthened pre-graduate training standards (from 2024/25) to increase practical exposure, communication skills, and cancer-care competencies. In 2020, they also modernised the post-graduate pathways to align with Directive 2005/36/EC. To accelerate entry, medical and dental students no longer have to wait until after graduation to take their final licensing exam, and trainees may take the national specialty exam in their final training year.

Summary

Poland uses a legally mandated, data-driven Map of Health Needs that links annual workforce evaluations with national and regional planning, education, and training to ensure a consistent and forward-looking approach to staffing

BOX 28 - Institutionalised evaluation through the Romanian national health institutes

How they evaluate

Romania evaluates staffing levels through a structured and institutionalised framework, anchored in its national public health institutions. The process is led by the National Institute of Health Services Management (NIHSM) and the National Institute of Public Health (NIPH), both of which play complementary roles in workforce planning and monitoring. The NIHSM develops tools for projecting future workforce needs and assessing service provision capacity, while the NIPH conducts evaluations of population health indicators to determine staffing requirements across the health system. Together, these institutions ensure that workforce evaluation is not only a technical exercise but also directly linked to health service delivery and population health priorities.

How they integrate the evaluation

Romania integrates staffing evaluation into broader policy frameworks, particularly the National Health Strategy and the Regional Health Services Master Plans. These strategies and plans are accompanied by monitoring and evaluation tools that allow policymakers to track progress, identify gaps, and adapt interventions. By combining health system data, epidemiological trends, and service delivery indicators, Romania has created a feedback loop that connects staffing assessments to long-term planning and policy implementation.

Summary

The Romanian approach emphasises institutional capacity and continuity as it embeds evaluation within permanent national institutions, giving it stability and credibility. This structure enables ongoing adjustments and ensures that staffing evaluations are consistently updated in line with health system reforms and demographic changes. While implementation challenges remain, specifically surrounding data integration and regional variation, the system provides a solid basis for evidence-based workforce planning, aligning staffing levels with national health priorities and regional service delivery needs.

BOX 29 - Evaluating Safe Nurse Staffing in Ireland

Ireland's Framework for Safe Nurse Staffing and Skill Mix was evaluated through a comprehensive pilot implemented across six wards in three hospitals. The setting for the research was all medical, surgical and specialist wards in the three pilot hospitals (excluding intensive care units, emergency and outpatients' departments). The central objective was to determine whether evidence-based staffing levels improved patient outcomes, strengthened the workforce, and generated economic value. The evaluation included clinical, organisational, workforce, and a full economic analysis.

Quantifying the costs of implementation

Introducing safe staffing required both a direct investment, i.e. hiring staff, and the conversion of agency staff to substantive posts, i.e. shifting staff from costly agency contracts into permanent or directly employed positions. The pilot identified the need for 21.2 whole-time equivalents (WTE) through new staffing investment alongside 15.1 WTE through agency conversion, producing a total requirement of 36.3 WTE across the pilot wards. The average investment per ward was approximately €160,000.

Initially, introducing safe staffing generated net additional costs because permanent staff were recruited before agency reliance had substantially decreased. Over time, however, sustained conversion of agency hours into substantive roles reduced dependence on costly temporary staffing. As these efficiency gains accumulated, the framework shifted from a net cost to generating net savings, demonstrating that early investment in stable staffing can produce long-term financial benefits.

Quantifying the benefits

The framework produced substantial, measurable improvements in both patient and staff outcomes.

Reductions in nurse-sensitive outcomes (NSOs), which are patient outcomes directly influenced by the quality and adequacy of nursing care, led to measurable improvements in both economic efficiency and overall care quality. Care left undone fell sharply: the proportion of nurses reporting at least one missed care item dropped from 75.6% to 31.8%, and the average number of missed items per shift fell from 2.51 to 0.75. Delays in care also declined, both in frequency and volume.

Workforce benefits included improved perceptions of staffing adequacy, stronger leadership and teamwork, and lower intentions to leave in wards receiving staffing uplifts.

Contribution to health system performance

The pilot showed that a structured methodology for determining safe nursing staffing levels enhances system performance by improving patient outcomes, boosting workforce stability and increasing efficiency, particularly through reduced agency use. These findings supported national rollout and demonstrated that safe staffing functions as an “invest to save” model that strengthens care quality and long-term sustainability. This example also stresses that a pilot can be an important first step in evaluating the feasibility, effectiveness and impacts of an intervention before its wider implementation in a specific context of the health system.

BOX 30 - Evaluating minimum nurse staffing regulations in Germany

Germany introduced a mandatory system of minimum nurse-to-patient staffing ratios across hospital settings, aiming to improve safety and reduce understaffing in high-risk units. Since 2019, these legally binding ratios apply to “nursing-intensive units,” including surgical wards, paediatrics, cardiology, neurology, and intensive care, covering 93.5% of all hospitalisation days. Non-compliance leads to financial sanctions, creating strong incentives for hospitals to meet the regulated thresholds. Examples include a 2.5:1 day-shift ratio in adult ICUs and 10:1 in internal medicine during daytime hours.

Evaluation approach

The system was formally evaluated by the National Association of Statutory Health Insurance Funds and the German Hospital Association. The 2024 evaluation covered 1,300 hospitals, analysing (1) compliance data and (2) interviews with Chief Nursing Officers to assess operational and workforce effects.

Quantitative and qualitative findings

The data analysis found that 15% of all shifts between 2021 and 2023 were understaffed. The most affected areas were special paediatrics (27%) and stroke units (19.9%), while general paediatrics (10.4%) and internal medicine (~11%) had the lowest levels of understaffing. Understaffing occurred at similar rates across day and night shifts.

Qualitative findings reveal significant operational consequences. 62% of hospitals reorganised services to comply with minimum ratios, and 61.9% reported blocking beds to avoid violations and associated penalties. Despite meeting ratios, 80.4% of nursing leaders reported no improvement in working conditions, suggesting that staffing thresholds did not fully align with actual workload or patient complexity. However, 42.3% reported an improved perceived feeling of safety among nurses, indicating some benefit for staff confidence and risk management.

Impact on health system performance

Germany's evaluation does not provide quantified financial benefits or cost offsets, meaning there is not enough information to determine whether minimum ratios improved system-wide efficiency or produced economic gains. The findings instead highlight unintended consequences, especially reduced bed capacity, and a perceived mismatch between mandated ratios and real care needs. This can help policymakers understand trade-offs between access to healthcare and measures aiming at safety regulation.

Germany is now transitioning toward PPR 2.0, a more flexible acuity-based staffing tool, currently regulated for adult and paediatric wards, with ICU-specific tools (INPULS) and doctor staffing instruments under evaluation. This shift reflects a policy response to the limitations identified in the evaluation, stressing its key role as a learning tool and a guide to refine health workforce policies, rather than a tool for the narrow assessment of compliance or performance alone.

BOX 31 - Expert-based and emerging approaches to staffing level evaluations in Bulgaria

How they evaluate now

Staffing level evaluations in Bulgaria are led primarily by the Ministry of Health together with the National Centre for Public Health and Analysis. Their role is to monitor and assess the availability of health personnel, paying particular attention to those specialised fields where shortages are most acute. Professional organisations, such as medical and nursing associations, also contribute their expertise and ensure that the evaluations reflect the realities of clinical practice. However, there is currently no standardised methodology for this assessment. Instead, evaluations are carried out largely at an expert-opinion level, relying on the knowledge and judgment of specialists rather than systematic or data-driven models.

What they plan to change

This ad-hoc and unregulated approach has led to inconsistencies in staffing assessments across different institutions and actors. Discrepancies between evaluations often make it difficult to establish a shared and reliable picture of workforce needs. Policymakers and stakeholders have acknowledged that this fragmented system undermines the capacity to plan effectively for shortages and imbalances in the health workforce. Recognising these limitations, the Bulgarian government has committed to developing a formal methodology for monitoring and evaluating its healthcare personnel. This reform is embedded within the country's Recovery and Resilience Plan, adopted by the Council of Ministers, which sets out a roadmap for building a more structured and evidence-based system of workforce planning.

Summary

By moving from expert-driven monitoring to structured methodologies, Bulgaria aims to improve accuracy, comparability, and reliability in its staffing assessments, aligning its workforce planning with broader European efforts toward evidence-based health system governance.

Sectoral level

Do countries see opportunities and risks of setting staffing levels at the sectoral level?

In hospitals, opportunities cluster around safety, transparency, and better funding alignment

Ireland's research-based *Framework for Safe Nurse Staffing and Skill Mix* shows measurable gains for patients and staff when ratios match the case-mix and acuity and are then evaluated independently (Box 32). Luxembourg's biennial process ties standards to budget cycles and mixes fixed and flexible norms by unit, improving fit to activity. Germany is rolling out a workload-based instrument – data collection is in process at the time of drafting this report to set requirements dynamically (Box 35), while Slovakia is moving to explicit maximum patient-to-staff ratios by department. Austria is re-evaluating special-unit norms and updating forecasts (via the National Public Health Institute) to match training pipelines.

The risks countries have identified in the hospital sector are not risks to staffing levels themselves but rather risks of how staffing levels are set. Czechia and Slovakia, for example, point to the danger of using outdated or rigid ratios that fail to reflect patient acuity or current hospital practices. In addition, Bulgaria highlights the inconsistencies and lack of reliability that arise when evaluations are carried out without a standardised methodology.

A big opportunity in primary and ambulatory care is the distribution of health and care workers

A trend is moving towards using minimum/optimal network designs and forecasting to put teams where patients are. Slovakia, for example, is mapping minimum/optimal regional networks for specialists and practices; Austria's GÖG model spans outpatient and inpatient flows and is being embedded in binding regional structural plans (Box 33); Hungary's monitoring system already tracks primary-care doctors and nurses and will feed a unified model that blends workload distribution with operational minimums; Portugal applies the same supply–demand tooling system-wide; and Italy's National Agency for Health Services is developing methodologies for defining professional needs.

The risks are clear. If methodologies are missing or fragmented, sectoral targets can harden inequities or misallocate posts. Even with good models, changing competencies can make headcount rules obsolete, so country standards would do well to evolve with the new roles and pathways.

Countries tend to see sectoral staffing levels as a chance to improve safety, transparency, and the alignment of resources with demand. At the same time, many acknowledge that without robust methodologies and adaptability, sector-specific rules risk becoming outdated, inconsistent, or unable to respond to changing workforce and patient needs.

BOX 32 - Ireland's research-based framework for safe nurse staffing and skill mix

What is it

Ireland has developed a very research-driven approach to hospital staffing in Europe through its Framework for Safe Nurse Staffing and Skill Mix. The Framework was created via a multi-year program of research, with independent evaluations carried out throughout its development. These evaluations demonstrated significant improvements in patient safety, quality of care, and staff well-being when the model was applied. Based on this evidence, the Framework was adopted as national policy in 2020, covering adult medical and surgical care settings, with implementation expanding to emergency care. The model is structured around nursing hours per patient, allowing ratios to be adapted to the complexity and acuity of patients across specialist areas, including critical care. While considered a fixed staffing approach for nurses, it has built-in flexibility, with different ratios set by care area to better reflect patient needs.

Why does it work so well

Staffing models aligned with patient needs tend to lead not only to better health outcomes but also to increased retention and improved job satisfaction for nurses. The Frameworks objectives explicitly include ensuring effectiveness, patient-centredness, safety of care, and protection of health workers' well-being. Its implementation is supported by system-level modelling approaches that project aggregated demand and supply across the public, voluntary, and private sectors, while also considering population change, healthy ageing, prevention policies, and task-sharing reforms. This shift away from density benchmarks towards a needs-based modelling approach positions Ireland as a frontrunner in linking workforce planning to future demographic and epidemiological realities.

Where do the opportunities lie

Ireland will extend the Framework to long-term residential care for older people and community/primary care. The model is being independently re-evaluated to test its ongoing effectiveness, while compliance monitoring systems are being developed to ensure consistent implementation nationwide. Though the Framework is not yet solidified by legislation, it is embedded in the national policy and represents a hybrid approach: fixed staffing ratios informed by rigorous research but applied flexibly across care settings. This strategy offers both predictability and adaptability, linking day-to-day staffing practices with broader workforce planning and training pipelines.

Summary

Ireland's Framework for Safe Nurse Staffing and Skill Mix uses a research-based, needs-driven model that improves patient safety and nurse well-being while guiding national workforce planning.

BOX 33 – Austria’s structural plan for primary care staffing is a balance between flexibility and consistency

What is it

Austria evaluates and plans staffing levels in primary care through the Austrian Structural Plan for Healthcare (ÖSG), which serves as the national framework for the entire public health system. The ÖSG sets binding guidelines for regional and sectoral planning, covering outpatient structures such as primary health care units (PHCUs), group practices, contracted physicians, and multidisciplinary teams. For doctors and nurses, staffing levels follow fixed ratios, while for other professions active in PHCUs, i.e. dietitians, physiotherapists, occupational therapists, speech therapists, psychologists, psychotherapists, social workers, and midwives, the model is more flexible, allowing adaptation to regional needs and patient demand. These requirements were adopted by the federal states through Regional Structural Plans for Healthcare (RSGs), which means that the national framework is now a binding regional workforce plan.

What makes it special

A central feature of Austria’s approach is the integration of population profiling, workload analysis, epidemiological data, and accessibility thresholds into planning. The ÖSG and RSGs define the required physician density per specialty, travel-time thresholds for accessibility, and criteria for quality of care. The National Public Health Institute (GÖG) supports this with a monitoring and forecasting model that uses pseudonymised individual-level data to project medical supply and demand across both inpatient and outpatient care. Updated annually, the model not only maps the current workforce but also forecasts retirements, training needs, and likely shifts between sectors. These insights are meant to guide decision-making on training places, redistribution of professionals, and adaptation of competencies.

Where do the opportunities lie

The aim of the model is to strengthen evidence-based, regionally sensitive planning while ensuring national consistency. By linking staffing levels to both health outcomes and workforce sustainability, Austria seeks to balance effectiveness, equity of access, quality of care, and staff well-being.

Summary

Austria’s Structural Plan for Healthcare combines fixed staffing ratios with flexible multidisciplinary planning, linking workforce needs to accessibility, quality, and long-term sustainability in primary care.

Do countries evaluate their staffing levels at the sectoral level?

At the sectoral level, several countries have established hospital-specific evaluations, many of which are aimed at ensuring patient safety and aligning staffing with service demand. Slovakia's Institute for Health Care Analysis monitors deficits across departments at both the hospital and national levels, while Bulgaria's Ministry of Health and National Centre for Public Health and Analysis track personnel availability in hospitals, although they do not yet have a standardised methodology. Austria is reforming hospital staffing standards, with a focus on special units such as palliative care, psychiatry, stroke units, and ICUs, supported by forecasting models developed by the National Public Health Institute (GÖG). Ireland provides an example of evidence-based evaluation through its Framework for Safe Nurse Staffing and Skill Mix, which is subject to independent research and has been integrated into national policy across surgical and medical hospital settings. Greece also evaluates hospital workforce needs through operational indicators like bed occupancy and hospitalisation rates (Box 34), while Luxembourg carries out biennial reviews of hospital staffing needs during budget negotiations. In countries like Czechia and Germany, reform discussions and surveys highlight the perception that hospital staffing frameworks are outdated, pointing to an evolving but not yet consolidated evaluation practice (Box 35).

Primary care evaluations tend to focus on ensuring adequate distribution and accessibility of health professionals across regions. Slovakia is developing models to define minimum and optimal regional networks for specialists and ambulatory care practices, thereby tailoring workforce availability to local patient needs. Portugal mirrors its system-wide approach by extending evaluation tools to the primary care sector, while Austria's GÖG forecasting model captures both outpatient and inpatient medical supply, offering an integrated view of workforce dynamics across sectors. Hungary uses its human resources monitoring system to generate trend analyses and forecasts that explicitly include primary care doctors and nurses, with attention to issues such as workforce ageing and outward migration (Box 36). Italy has delegated responsibility to Agenas (the National Agency for Health Services), which is actively engaged in workforce planning for the primary care sector. Bulgaria follows a less formalised approach, monitoring availability through expert-level assessment without a dedicated methodology. These sectoral evaluations highlight the diverse strategies countries use to adapt workforce planning to the distinct requirements of primary care, which often emphasise accessibility and continuity of care over the acute service pressures typical of hospitals.

Many of the surveyed countries evaluate their staffing levels at the sectoral level, particularly in hospital and primary care. Hospital evaluations often focus on patient safety, service demand, and operational indicators, while primary care evaluations emphasise equitable distribution and accessibility.

BOX 34 - Evaluating hospital staffing levels in Greece

How they evaluate

Greece runs a centrally coordinated, indicator-based process of evaluation for hospitals. Each year the Ministry of the Interior and Ministry of Finance set the total number of posts that can be announced. Public hospitals can then submit requests to Regional Health Authorities (RHAs). Using operational indicators, the Ministry of Health allocates approved posts across RHAs and hospitals. Doctors can also be hired as auxiliary or contracted staff (typically for 12–24 months) to allow for flexibility between recruitment rounds. Nursing and other personnel are recruited permanently via the Supreme Council for Civil Personnel Selection, with the Ministry of Health allocating posts as they are needed.

What evidence is used

Allocation decisions consider a common set of service-pressure metrics: beds, bed occupancy, number of clinics, days of hospitalisation, hospitalisation rates, surgeries per year, and other operational indicators. These are combined with population profiling, workload/service-use data, local workforce profiling (distribution gaps), and modelling of organisational constraints (i.e. approved organograms, expected retirements). The approach is flexible by design thereby aims to best reflect Greece's geography, especially the islands and the mountainous areas, where access and recruitment challenges require tailored responses.

How do they consider the dispersed geography

To staff remote or underserved areas, Greece provides monthly financial incentives, relocation and housing support, faster career progression, extra education leave, and other non-financial benefits. Bonuses are targeted at specialties such as internal medicine, family medicine, anaesthesiology. Compliance is monitored via regular reporting and audits (vacant/filled posts, re-advertisements), BI-Health (Business intelligence for health) submissions from hospitals to RHAs, and a central auxiliary personnel registry within the Ministry of Health.

Digitalisation and continued improvement

A new Human Resources Information System (HRIS), a part of the Quality for All / HEALTH-IQ program, consolidates data and indicators to track both quantitative staffing and skill-mix. The first operable version was planned for October 2025, and the full delivery will be effective for February 2026. Together with legally binding post limits and indicator-driven allocation, this would tighten the feedback loop between needs, recruitment, incentives, and regional distribution, allowing Greece to keep hospital staffing aligned with effectiveness, safety, access, workforce well-being, and retention goals.

Summary

Greece uses an indicator-driven, centrally coordinated process to allocate hospital staff, combining operational metrics with geographic incentives and a new HRIS to align recruitment with system needs and workforce sustainability.

BOX 35 - From fixed ratios to evidence-based evaluation in Germany

How they evaluate

Germany evaluates hospital staffing levels through a mixed regulatory and research-based framework. For nurses, the Federal Ministry of Health introduced the Pflegepersonaluntergrenzenverordnung (PpUGV), which sets fixed minimum staffing levels in key hospital departments such as general care, intensive care, surgery, maternity, and

psychiatry. Hospitals are legally required to provide quarterly reports on compliance with these ratios, including the number of shifts that fell below minimum levels. Persistent non-compliance triggers financial sanctions. For psychiatric care, the Federal Joint Committee (Gemeinsamer Bundesausschuss, G-BA) has developed the Personalausstattung Psychiatrie und Psychosomatik-Richtlinie (PPP-RL), which uses acuity-based assessments tailored to patient needs.

What they are changing

While minimum ratios currently dominate, Germany is transitioning towards workload-based evaluation tools. At the same time, a federal regulation from July 2024 requires hospitals to submit data that will underpin a new nurse staffing measurement system (Pflegepersonalbemessungsverordnung, PPBV), moving away from rigid ratios towards more adaptive, workload-driven planning.

Why it is important

Germany's model balances legal minimum guarantees with a gradual shift toward evidence-based planning. Current ratios ensure a transparent floor that cannot be undercut, making risks of understaffing highly visible and sanctionable. However, the system does not yet allow flexible adjustment to local hospital realities, except in extraordinary cases like pandemics. Germany is moving from compliance-driven evaluation toward a dynamic, data-informed approach that seeks to align staffing with patient needs, staff well-being, and system sustainability.

Summary

Germany combines legally mandated minimum staffing ratios with new workload-based evaluation tools, moving from compliance-driven regulation toward a more flexible, evidence-based approach to hospital staffing.

BOX 36 - Hungary's centralised evaluation of primary care staffing

How they evaluate primary care

The evaluation of primary care (PC) staffing is centralised and statute-based. Two national bodies lead it: the National Hospital Directorate General (OKFŐ) and the National Centre for Public Health and Pharmacy (NNGYK). Under the Health Act, the Minister responsible for Health (currently the Minister of Interior) sets sectoral policy, defines minimum professional conditions for service provision, and issues the detailed rules that govern provider organisation and operation. OKFŐ also acts as the organiser for family medicine districts ("praxiskezelő"), establishing practice areas and ensuring equal territorial access.

OKFŐ operates a sectoral HR monitoring system that integrates operational records of health workers, payroll and staffing statistics, PC workforce data (GPs, paediatric GPs, dentists, nurses/health visitors), and indicators of outward mobility. Regular reports, trend analyses, and forecasts are produced, i.e., expected nurse exits due to ageing, or the impact of wage measures. Planning combines population profiling (demographics and geography), workload and utilisation data (service use, local demand), workforce profiling (distribution gaps), and modelling of organisational constraints (approved posts, infrastructure). Minimum catchment thresholds anchor the service design: at least 600 residents for paediatric GP districts and 1,200 residents for adult/mixed GP districts, with adjustments for local characteristics to guarantee access and efficiency.

Updated competency frameworks

Alongside minimums, Hungary is updating competency frameworks: a new decree clarified nursing competency levels (2025), and work is underway to refine the competencies and powers of family physicians and family paediatricians, supporting task sharing and more efficient team mixes.

The link to investment and financing

Evaluation is tied to financing and investment. NEAK (the payer) funds wage components and provides incentives for GP clusters/practice communities that enable shared staffing and flexibility, with extra support for disadvantaged areas. Findings from the monitoring system feed into training quotas (i.e., state-funded specialty training or nursing education), wage policies, and targeted scholarships thereby aligning supply pipelines with observed local and sectoral needs.

Summary

The rationale for Hungary's approach is pragmatic. The fixed minimum thresholds provide a clear, auditable baseline for institutional HR planning and budgeting. District-level capacity control adapts to social need while evolving competency rules keep the model responsive to the changing roles in primary care teams.

Conclusions: What considerations do you have to think about when implementing or changing staffing levels?

Implementing or changing staffing levels is not simply a technical exercise, rather it is a policy decision that requires balancing evidence, flexibility, and governance. While staffing ratios can seem like a clear-cut solution, the reality is that their effectiveness depends on context, the methods used to define them, and how they are embedded in a country's wider health system goals and health workforce policies.

Ratios are not transferable, but methodologies are

A key consideration is that numbers themselves are not transferable across countries; however, the methodologies behind them are. Ratios that work in one country may be unsuitable in another due to differences in workforce composition, service delivery models, or financing structures. What can be shared are the approaches for determining those ratios, i.e. the analytical methods that underpin them. Tools that calculate workload, assess patient acuity, or model workforce demand can be adapted to different systems and used to support international learning without reproducing raw figures or inappropriate benchmarks.

Existing staffing models can be adapted, they do not have to be replaced

Staffing models do not always need to start from scratch. Long-standing frameworks can evolve through evidence-based review rather than being completely replaced. Continuous evaluation and adjustment are key to ensuring that staffing levels remain fit for purpose, realistic and credible. This is important because they are then better able to reflect the changes that are likely to arise in service delivery, patient needs, and workforce capacity.

Not defining ratios still means someone needs to make a decision

A system without defined staffing levels does not mean staffing decisions disappear, it simply means they are made flexibly without clearly identifying decision makers. In such systems, decisions tend to be made at the local, institutional or managerial level. This can work well in some contexts but may increase the risk of inconsistency and inequity across facilities or regions. In effect, a lack of a formal definition simply transfers decision-making power to a less transparent or less regulated level.

Different levels of governance play distinct roles

Governance is another critical dimension. In many countries, staffing levels are determined at the regional or local levels, with the national authorities focusing on workforce planning, forecasting, and strategic coordination. This division of roles allows flexibility close to service delivery while maintaining alignment with national objectives. It also highlights the importance of coherence between the levels: if national forecasting is disconnected from local implementation, mismatches between supply, demand, and funding can quickly emerge. However, not all countries define staffing levels explicitly, and thus there is often a fear that doing so might reduce the quality or flexibility of care. Some governments and professional bodies often worry that fixed ratios could be interpreted as maximums rather than minimums, leading to rigid application and a potential decline in care standards. Addressing these concerns requires adaptive approaches that pair defined minimums with mechanisms that preserve professional judgment and allow for contextual adjustments.

Bureaucracy can stifle useful reforms

Overly complex reporting requirements or slow approval processes can delay implementation and discourage innovation. Streamlining governance processes and focusing on practical, evidence-based evaluation is essential to make staffing reforms effective rather than burdensome. Simplified yet transparent accountability mechanisms enable systems to evolve while maintaining oversight.

In the same vein, extensive stakeholder consultations can slow down reforms. In some countries, governments must consult a wide range of stakeholders before deciding on staffing levels or releasing data for international comparison. While broad consultation is critical to ensure ownership and legitimacy of staffing policies, overly complex or lengthy stakeholder processes can hinder timely decision-making. Streamlined, transparent engagement mechanisms can help maintain momentum while preserving trust.

Staffing reforms must connect health system goals with health workforce goals

Effective implementation ultimately requires a clear link between health system goals and health workforce goals. Concentrating only on ones without the others risks creating distortions: a well-staffed system that fails to meet population health needs, or a well-designed service model with too few staff to deliver it. Evaluating staffing reforms through both lenses ensures that workforce capacity supports system performance, and that system design makes efficient and sustainable use of human resources.

In summary, successful practices with staffing ratios are those that are context-sensitive, evidence-informed, and grounded in continuous evaluation. They rely less on copying ratios and more on adapting methodologies, clarifying governance, and integrating workforce and system objectives. The challenge is to create frameworks that are both structured and flexible – providing assurance where it is needed, but leaving space for adaptation, innovation, and learning across countries.

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ANNEXES

Annex 1 - Main conclusions of the Policy Focus Group meeting

The policy focus group plays a critical role in supporting the work of the HSPA EG. It ensures that performance measurement is aligned with relevant policy priorities, facilitating dialogue between policymakers, technical experts, and stakeholders. Specifically, the policy focus group helps interpret performance data through a policy lens, identifying the implications of results for system governance, resource allocation, workforce planning, and service delivery reforms. In addition, the policy focus group can contribute to setting evaluation agendas, prioritising areas where performance gaps have significant policy relevance, and guiding the development of recommendations that are both evidence-based and politically viable. Through these functions, the policy focus group strengthens the link between performance assessment and effective policy action, enhancing the overall responsiveness and accountability of the health system.

The findings and main conclusions of the survey on staffing ratios were discussed with stakeholders representing health professionals and healthcare providers. The discussion followed a presentation by the European Observatory on Health Systems and Policies at the dedicated policy focus group meeting, which took place on 5 November 2025 at the invitation of the HSPA EG co-Chairs. One organisation could not be present and shared written remarks that are also taken into account here.

Participants widely appreciated the efforts of the HSPA Expert Group in addressing this critical issue and expressed confidence that the report would significantly contribute to discussions at multiple levels. By complementing the perspectives and research conducted by health professional organisations, the report is expected to enhance focus on staffing ratios and stimulate debate on adaptable approaches ensuring that staffing ratios are tailored to specific healthcare contexts and health systems.

The conclusions from the policy group discussion emphasised some key considerations for future focus:

How to ensure that staffing ratios are fit for purpose?

The meeting centred around refining the understanding and implementation of safe staffing ratios within healthcare systems. Participants stressed the importance of stakeholder collaboration, with emphasis on balancing flexible and fixed staffing models tailored to specific healthcare contexts and dynamic developments in relation to demand for healthcare and modernisation of health systems. However, some healthcare professional representatives stressed that frontline healthcare professionals benefit greatly from fixed safe staffing ratios, as they offer a transparent and easily understood framework which leads to safer workplaces and improved quality of care. The distinct variations in national healthcare systems necessitate adaptable approaches that consider regional and local disparities and evolving policy priorities (possibly reflected in health system reforms), ensuring alignment with both health system goals and broader societal objectives. While implementing universal benchmarks across the EU could be challenging due to inherent variability among Member States, an EU framework on safe staffing levels could support convergence towards more equal healthcare standards between Member States. Furthermore, the efforts of the Expert Group on HSPA (HSPA EG) offer valuable resources for facilitating experience-sharing between countries and comparing the perspectives of Ministries and health professional organisations.

Evaluation of existing staffing level solutions can lead to more insightful discussion

The dialogue underscored the need for evaluation to discern the effects, benefits, and challenges associated with staffing ratios. Evaluations help identifying disparities and guiding targeted reforms, although the survey indicated they are still underused for this purpose. The discourse emphasised that staffing levels must be complemented with broader strategies concerning the health workforce and working conditions to improve their effectiveness. Furthermore, a stronger connection with health workforce planning can position staffing ratios as a key element of health system reforms, taking into account a more dynamic perspective of changing needs and evolution of care models.

Engagement and dissemination

Participants committed to disseminating the insights from the report and will welcome opportunities of further discussion at EU level. This will complement dissemination activities by the HSPA EG. Plans include sharing findings with national governments and leveraging the report for continuous policy dialogues with stakeholder organisations. The report acts as an important mapping exercise, complementing the work conducted by the healthcare professionals associations, offering essential data to Member States and highlighting the necessity for implementing concrete solutions in workforce planning for the benefit of frontline healthcare professionals and patients. The importance of conducting a nuanced analysis that addresses practical challenges and facilitates effective communication with policymakers was strongly emphasised. The dissemination efforts aim to extend the reach of the findings, enabling national experts to facilitate dialogues within their respective ministries, and support a thoughtful examination towards the implementation of staffing levels in the context of health system reforms and workforce sustainability.

Looking ahead

The importance of implementation, and dissemination was echoed throughout the meeting, with future opportunities to support widespread access and discussion of the report's findings. The meeting underscored the significance of collaborative implementation research efforts involving European and National Healthcare Professionals Associations and Member States and highlighted a shared commitment to developing concrete solutions that address real-world challenges within healthcare workforce planning. The session also stressed the necessity of integrating staffing ratios discussions into broader workforce strategies that focus on sustainability and resilience in healthcare systems.

List of participants to the Focus Group

- [CPME](#) (Standing Committee of European Doctors),
- [EFN](#) (European Federation of Nurses Associations),
- [UEMO](#) (European Union of General Practitioners),
- [HOSPEEM](#) (European Hospital and Healthcare Employers' Association),
- [HOPE](#) (European Hospital and Healthcare Federation),
- [EHMA](#) (European Health Management Association),
- [EPSU](#) (European Public Service Union),
- Country representatives of the HSPA working group on healthcare workforce: Cyprus, Ireland, Poland

Annex 2 – Survey questionnaires

Survey I System level

Survey II Sectoral level

Staffing levels in European health systems: Why are they used, how are they defined, where are they implemented?

1

A survey co-designed by the European Observatory
on health systems and policies and the HSPA Expert Group
– working group on healthcare workforce

Defined staffing levels at the HEALTH-SYSTEM LEVEL



TABLE OF CONTENTS

Introduction	3
Section one: Approaches to and rationale for staffing levels	5
Section two: Governance	12
Section three: Evaluations, recent and planned reforms	20

INTRODUCTION

Background to the questionnaire

High-performing health systems rely on having the right number and mix of health workers in place to achieve health system goals and objectives, such as those related to health outcomes, staff and patient satisfaction, access, and quality. Shortages or imbalances in staffing can lead to delays, unsafe treatments, and poor user experiences, especially in ambulatory care clinics, emergency departments, and primary care settings to name a few. In the frame of Health System Performance Assessments (HSPA), staffing levels can be used as a set of representative measures to link workforce availability to broader health system objectives (i.e. access to healthcare, quality and safety of care, and population health, to name a few).

As countries embark on Health System Performance Assessments (HSPA) it would be beneficial to identify one or several indicators that help systems understand the link between workforce availability and broader health system objectives, and highlight areas that require further investigation.

Currently, many European countries have or are seeking to define and implement staffing standards across various sectors of their health systems. However, the terminologies and approaches used to establish these standards may differ between (and within) countries and there is little systematic knowledge about what countries are doing or about the emerging trends in staffing-related reforms.

The HSPA Expert Group identified health workforce as a priority topic for 2024-2025. After discussions and presentations on the EU actions addressing health workforce, the Expert Group decided at the October 2024 plenary to carry out a survey on existing practices regarding staffing levels, and their impact on the policies addressing health workforce shortages, to organise a workshop and issue a conference report. At the request of the European Commission, the European Observatory on Health Systems and Policies, an observer in the Expert Group was requested to support the work.

Survey aim

The aim of this survey is to therefore map approaches and methods to address staffing levels in the healthcare workforce in EU Member States and in Norway. By understanding how staffing levels are defined, implemented, utilized and evaluated across countries, there is an opportunity to facilitate mutual learning and identify best practice cases. The survey has been developed following several rounds of consultation with the HSPA co-chairs and members of the HSPA working group on the healthcare workforce. It was emphasized that the survey should map different healthcare sectors (e.g. hospital, primary care, etc.) and distinguish between doctors and nurses, as well as consider other professions.

Survey instructions

This survey consists of **two parts**:

- **Part 1 (HEALTH SYSTEM)** is focused on defined **staffing levels at the health-system level** in EU Member States and in Norway. Part 1 should be completed before Part 2.
- **Part 2 (SECTORAL)** is focused on defined staffing levels in **specific healthcare sectors** such as hospitals, primary care, ambulatory care etc. Part 2 should be completed if staffing levels at the sectoral level differ from those at the health system level.

This questionnaire is on **PART 1 (HEALTH SYSTEM)**.

Before answering questions in the three main sections, **please fill-in the box below**.

Please indicate for which country or region you are responding:



Please indicate your name and function:



We truly appreciate your time and thank you for sharing your expertise and experience. The analysis of the results will be part of a HSPA Expert Group report.

■ Section 1:

Approaches to and rationale for staffing levels

1.1a What focus and approach to staffing levels is used at the health system level in your country/region: fixed ratios, flexible staffing models, or none?

Instructions

Please indicate whether your country/region uses fixed staffing ratios, flexible staffing models or no defined staffing levels in your health system (see Box 1 for definitions) by checking the box next to the most appropriate answer(s) ☐. Staffing levels at the system level may, for example, take the form of: absolute numbers (of health workers or specific types of health professionals (doctors nurses etc)); ratios vis-à-vis population (health workers in total, or for different professions); ratios between professions (4 nurses per medical doctor etc). Please also check the appropriate profession(s) this approach applies to. Countries that adopt a combination of staffing ratios can also use the section marked “*further comments*” to elaborate on their answer by clicking on the text box and typing. If there are sources available, please include these in the box below labeled “*source(s)*”.

Box 1: Definitions

Countries adopt different approaches to defining staffing levels:

Fixed ratio staffing levels usually predefined ratios, typically specifying the number of health workers or professionals relative to patients, population groups, hospital beds, geographical areas... These ratios may be legally mandated or outlined in recommendations or guidelines.

Flexible staffing models are defined based on a combination of supply and demand factors aligning health worker availability with the healthcare needs of the population. This approach offers flexibility to adjust to workforce dynamics and evolving needs.

Other countries have no defined staffing levels. In some countries staffing levels are not formally defined, and neither fixed ratios nor flexible staffing models are utilized

Minimum threshold is the lowest number of healthcare workers that is deemed necessary to maintain safe, effective, and sustainable service delivery. This can include thresholds in the form of aspirational targets, guiding principles and mandatory measures.

Please select all answers that apply

☐ My country/region uses fixed staffing ratios at the health system level for:	☐ My country/region uses flexible staffing models at the health system level for:	☐ My country/region does not use defined staffing levels at the health system level
☐ Doctors	☐ Doctors	
☐ Nurses	☐ Nurses	
☐ Other professions - please specify here: 	☐ Other professions - please specify here: 	

Further comments:



Source(s), if available:



1.1b Does your country/region use minimum thresholds?

Although defined staffing levels for health systems may not be commonly established in Europe, methodologies to define minimum thresholds (see Box 1) for the number of health workers in a country/region do exist (Box 2).

Box 2: Methodologies to define minimum threshold for health workers in a country/region

WHO method

The resulting “SDG index threshold” of 4.45 doctors, nurses, and midwives per 1000 population was identified as an indicative minimum density representing the need for health workers.

<https://iris.who.int/bitstream/handle/10665/250330/9789241511407-eng.pdf>

IHME method

This method refers to the Institute for Health Metrics and Evaluation (IHME)'s approach to global health data collection, analysis, and modeling. This is the most recent methodology which has been developed and presented in the 2024 Health at a Glance Report. It points to thresholds of 3.54 doctors and 11.45 nurses and midwives per 1 000 population.

<https://www.sciencedirect.com/science/article/pii/S0140673622005323?via%3Dihub>

1.1b If your country/region uses minimum thresholds, have they been influenced by the international frameworks listed in Box 2, or otherwise based on other scientific evidence or based on a consensus among experts?

- ☐ WHO method
- ☐ IHME method
- ☐ Other scientific evidence, please specify alternative methods below
- ☐ Other expertise, please specify below
- ☐ Don't know
- ☐ My country/region does not use minimum thresholds

Specify other information, if applicable:



Source(s), if available:



➔ If your country/region does **not** define staffing levels at the health system level or use minimum thresholds, proceed to question 1.4 and provide reasons for why staffing levels have **not** been defined.

1.2 What are the reasons for defining system-wide staffing levels in your country/region?

Instructions

In Box 3 below several goals are defined. These can be the reasons for defining staffing levels in a country/region. We have identified three broad categories:

- Health system objectives and goals
- Health workers related goals
- Societal goals.

To answer the following questions, please refer to official documents such as policies, legislation, and guidelines. These may be issued by the government, the Ministry of Health, competent arm's length bodies, social partners delegated to draft such documents. If, for example, *effectiveness of care* is an official reason as to why your country/region is defining staffing levels, please check the box next to it ☐.

Several official goals can be reported. If the documents you have identified include additional reasons for defining staffing levels, please specify these in your response. Please include in writing or provide a weblink or an attachment for the sources you reference.

Please select all answers that apply

Health system objectives and goals	Goals related to health workers	Societal goals – well-being
☐ Effectiveness of care	☐ Protecting the health and well-being of health workers	☐ Social cohesion
☐ Safety of care	☐ Increasing retention by improving working conditions	☐ Economic development
☐ User experience	☐ Don't know	☐ Environmental sustainability
☐ Access to care		☐ Don't know
☐ Patient centeredness		
☐ Health improvement		
☐ Financial protection		
☐ Quality of care		
☐ Don't know		

Other relevant health system objectives and goals:



Other relevant health workers related goals:



Other societal goals:



Source(s), if available:



Box 3: Definition of declared objectives and goals for defining staffing levels

Health system objectives and goals: they include people-centeredness, health improvement, and financial protection. These goals are the outcomes used to measure the performance of a health system. They are very relevant to patients and citizens. The health workforce influences these final goals by impacting the intermediate objectives.

Effectiveness of care: The ability of an intervention to have a meaningful effect on patients in normal clinical conditions.

Safety of care: The prevention of errors and adverse effects associated with health services.

User experience: Users' beliefs, preferences, perceptions, responses, and behaviors that occur before, during and after.

Access to care: The opportunity or ease with which individuals or communities are able to use appropriate health services.

Patient centeredness: As a goal which captures how far the health system adequately addresses people's non-medical health needs.

Health improvement: A fundamental goal of any health system. It encompasses the health of the population at different points in the life cycle and is measured through morbidity and mortality indicators as well as aggregate life expectancy.

Financial protection: Refers here to the health system's role in protecting the population against risk. For example, if there are not enough doctors working in the public sector, people may be forced to visit a doctor in the private sector, leading to out-of-pocket spending.

Quality of care: is used as a generic term comprising various dimensions including e.g. access, safety, effectiveness, user experience but also efficiency and equity.

Goals related to health workers: These goals focus on safeguarding the well-being of health workers, ensuring their job satisfaction, and creating a supportive work environment that enables them to thrive in their roles.

Protecting the health and well-being of health workers: adequate staffing level are a key to ensure that health workers are not exposed to constant work overload affecting their health.

Improving retention by improving working conditions: adequate staffing levels are an instrument to retain health workers in healthcare organizations.

Societal Goals: These goals are outside the direct scope of the health system but are still closely linked to staffing levels. For example, the environmental impact of health workers' green skills can contribute to sustainability. A well-staffed health workforce can also positively affect economic development by supporting a healthier population, which in turn contributes to a stronger economy. Secure staffing levels during times of economic crisis have been shown to help maintain economic stability. Additionally, staffing levels can affect social cohesion, especially in areas with limited access to care, such as medical deserts or deprived urban areas.

Social cohesion: to achieve collective commitment to unity, mutual support, and shared responsibility among individuals and communities. This goal encompasses eliminating discrimination, reducing income inequality, and ensuring equal access to healthcare.

Economic development: is a vital societal goal which enables higher living standards, job creation, and increased innovation.

Environmental sustainability: refers to the interconnectedness of environmental concerns with broader sustainability objectives. It refers to a balanced and inclusive approach to economic growth that seeks to meet the needs of present and future generations without compromising natural resources and global ecosystems.

NOTE Intermediate Objectives:

These objectives are directly related to how well health systems deliver services and achieve overall health outcomes. They include effectiveness, safety, user experience, and access to care. Achieving these objectives is crucial for high-quality service delivery. For this to happen, health systems need the right number of skilled workers in the right places at the right times. These intermediate objectives are a key link between staffing levels and the final health system goals. If intermediate objectives aren't met, it's unlikely the system will perform well in terms of people-centered care, health improvement, and financial protection. The relationship between staffing levels and health system outcomes can be understood through these intermediate objectives.

1.3 What were the reasons for choosing one approach (e.g., fixed ratios or flexible staffing models) over the other?

Please provide the reasons for choosing one approach (e.g., fixed ratios or flexible staffing models) over the other. To guide your response, here are a couple of examples (see Box 4):

Box 4: Reasons for preferences – examples

- “We have chosen fixed ratios because it is much easier to define and it provides enough orientation for recruitment and health workforce development”.
- “Flexible staffing models are too complicated and necessitate too much research.”
- “Flexible staffing models take local variations into account and define much better the health workforce needed at local/provider level”.
- “A consensus among experts suggested it was the best approach”.
- “The choice was informed by scientific evidence”.

Why has your country/region chosen one approach to define staffing levels over the other?



Source(s), if available:



➞ Please now **go to section 2** – governance.

1.4 Why are system-wide staffing levels not defined?

Instructions

This is an open-ended question. Please provide reasons why system-wide staffing levels have not been defined in your country/region. To guide your response, here are a couple of examples (see Box 5):

Box 5: Examples for reasons not to define staffing levels

- "Staffing levels are defined at sectoral level (hospital, primary care), rather than system-wide".
- "The government was worried that 'one-size-fits-all' would harm already well-established individual staffing levels at local or provider level".
- "The government felt there is not enough evidence for supporting the definition of staffing levels".
- "Once staffing levels defined, those providers which exceed the levels may decrease their workforce".

If it is difficult to base your response on an official document, please use any relevant knowledge you can obtain and specify the sources you reference. If no sources are available, input from experts or stakeholder views are also welcome.

Reasons for NOT defining staffing levels



Source(s), if available:



➔ If system-wide staffing levels are **not defined** in your country/region, **please go to section 3.**

■ Section 2: Governance

In this section we would like to understand the process of defining and implementing staffing levels.

2.1 Who sets the system-wide staffing levels in your country/region?

Staffing levels can be set by the government or the Department/Ministry of Health. In some countries, this responsibility may also be delegated to a specialized authority. We would like to understand the roles and interactions of the various stakeholders involved in setting staffing levels for health systems in your country/region. These may include professional associations, local authorities, and in some cases expert advisory councils or social partners. Please provide details on how these entities collaborate or influence the decision-making process related to staffing levels in health systems.

Who sets the system-wide staffing levels in your country/region?



Source(s), if available:



2.2 What tools and methods are used to determine the system-wide staffing levels in your country/region?

When staffing levels are defined, scientific tools and methods are often used to determine supply and demand (see Box 6 for definitions). This can apply to both fixed ratios and flexible staffing models. **Please check the boxes below** that apply, or fill in what you mean by “other”.

Box 6: Description of terminology

Population profiling: the analysis of demographic, epidemiological, and socio-economic characteristics of a population to inform health workforce planning.

Health service usage/workload profiling: analyzes how healthcare services are utilized and the workload distribution among health workers. Helps define optimal staffing levels by ensuring that workforce supply matches service demand effectively.

Provider/local health workforce profiling: the assessment of the composition, distribution, and capacity of health workers within a specific geographic area or healthcare system. Ensures that staffing levels are aligned with local population health needs, service demand, and healthcare delivery capacity

Clinical guidelines: recommendations that outline best practices for defining staffing levels by determining the type, number, and mix of health workers required to deliver high-quality care efficiently.

Modelling approaches: refer to the use of mathematical, statistical, and computational techniques to estimate and predict staffing levels based on healthcare demand, workforce supply, and system dynamics. Help policymakers and planners make informed decisions about workforce distribution, recruitment, and training needs.

Protocols: standardized procedures or guidelines that help determine staffing levels by specifying the required number, skill mix, and distribution of professionals needed to deliver care efficiently and safely

Please select all answers that apply

- ☐ None
- ☐ Population profiling
- ☐ Health service usage/workload profiling
- ☐ Provider/local health workforce profiling
- ☐ Clinical guidelines
- ☐ Modelling approaches
- ☐ Protocols
- ☐ Other, please give details

Further comments:



Source(s), if available:



2.3 Are system-wide defined staffing levels in your country/region legally binding?

In some countries/regions staffing levels may be mandatory without a legally binding instrument, especially in national health services, where there is a clear line of command from the central government to local providers. In contrast, systems with private or independent providers often rely on legally binding regulations for mandatory defined staffing levels, although enforcement may not always be directly handled by the Ministry of Health. In such systems, enforcement could come through the courts, initiated by health workers, citizens, or local authorities. In health systems with strong social partnerships, staffing levels may be governed by contractual agreements that establish legally binding obligations. Other countries may opt for a more flexible approach, relying on transparency and good faith rather than legal enforcement.

Please specify if system-wide staffing levels in your country/region are legally binding:

- ☐ Yes
- ☐ No
- ☐ Don't know

Further comments (i.e. what is the legal status, what type of instrument are you using if staffing levels are not legally binding):



Source(s), if available:



2.4 How is compliance with system-wide staffing levels monitored?

We would like to understand how the implementation of system-wide staffing levels is monitored and who is responsible for doing so. Monitoring adherence to fixed staffing ratios is typically straightforward, as it involves tracking whether the established ratios are met. However, monitoring flexible staffing levels is more complicated, because it requires verifying that the rules are being correctly applied.

How is compliance with system-wide staffing levels monitored?



Source(s), if available:



2.5 To what extent can the regional or local healthcare provider level (hospital or other) adjust national staffing levels?

This question seeks to understand the role of the regional and local/provider-level authorities in determining staffing levels. There are often significant regional and local differences in both population demographics and the labour markets for healthcare workers. These differences can exist between urban and rural areas but also between affluent and underserved regions. We would like to know the extent to which regional or local/provider-level entities have the flexibility to adjust national staffing levels.

To what extent can the regional or local healthcare providers (hospital or other) adjust national staffing levels?



Source(s), if available:



2.6 System-wide defined staffing levels and health workforce planning: is there a connection, and if so, what does it look like?

Setting defined staffing levels can directly impact the number of health workers needed as well as the required skill-mix. Adjustments in staffing may require planning for an increase in the broader healthcare workforce or a focus on specific professions. We are interested in understanding whether there is a connection between staffing levels and health workforce planning in your country/region. For example, introducing minimum doctor-to-population ratios may require targeted incentives for recruitment and retention, or a reduction in other health professional staff. If such a connection exists, what mechanisms or processes are in place to align staffing levels with health workforce planning needs?

2.6a Does a connection between defined staffing levels and health workforce planning exist in your country/region?

- ☐ Yes
- ☐ No
- ☐ Don't know

➔ If you answered "no" or "don't know", please go to question 2.7.

2.6b If such a connection exists, what mechanisms or processes are in place to align staffing levels with health workforce planning needs?



Source(s), if available:



2.7 System-wide defined staffing levels and investment in the health workforce: is there a link, and if so, what does it look like?

Changes in staffing levels can also have significant implications for investment in the health workforce, as they may lead to a change in the number of workers or a shift in the required skill mix. This, in turn, could require additional financial resources for salaries, training, and other workforce-related investments. We would like to understand whether there is an established connection between staffing levels and investment in the health workforce. If such a connection exists, how is it reflected in budgeting or resource allocation for salaries, training, and other workforce needs?

2.7a Does a connection between defined system-wide staffing levels and health workforce investment exist in your country/region?

- ☐ Yes
- ☐ No
- ☐ Don't know

➞ If you answered "no" or "don't know", please go to question 2.8.

2.7b If such a connection exists, how is it reflected in budgeting or resource allocation for salaries, training, and other workforce needs?



Source(s), if available:



2.8 System-wide defined staffing levels and training capacities: is there a link, and if so, what does it look like?

Changes in staffing levels may impact the need for training capacities at universities and vocational training institutions. As the demand for healthcare workers fluctuates, training programs must be adjusted to ensure alignment with these needs. This could involve close collaboration with various ministries, such as those in charge of science, education, as well as with universities, medical schools, and other professional training institutions. We are interested in understanding whether there is a connection between staffing levels and training capacities, and how this connection is managed across different sectors and institutions.

2.8a Does a connection between defined system-wide staffing levels and health workforce training exist in your country/region?

- ☐ Yes
- ☐ No
- ☐ Don't know

➞ If you answered "no" or "don't know", please go to section 3.

2.8b What type of connection between system-wide defined staffing levels and health workforce training exists?



Source(s), if available:



Section 3:

Evaluations, recent and planned reforms

In this section, we ask you to identify existing evaluations, recent reforms and future plans or policies related to system-wide health workforce staffing levels in your country/region. For **evaluations**, please consider highlighting any evaluation methods used by your country/region (see Box 7 for examples). For **recent reforms**, please consider highlighting any significant changes, updates, or new policies implemented in the last 5–10 years that impact staffing levels (see Box 8 for examples). For **future plans or policies**, please describe any planned initiatives, strategies, or policies under development aimed at addressing staffing levels (see Box 9 for examples).

Box 7: Examples for existing evaluations

- “The *National Institute for xx* has developed workforce planning tools that evaluates patient acuity and dependency to determine staffing levels.”
- “*Institute xyz in country abc* evaluates population health to determine staffing requirements in the health system.”

3.1 Please identify any existing evaluations of system-wide health workforce staffing levels in your country/region:



Source(s), if available:



Box 8: Examples for recent reforms

- “My country implemented a legislation introducing fixed staffing ratios to determine staffing levels in health systems in 20...”
- “In 2020, my country, has reformed the Framework Statute for Health Personnel, setting limits on temporary work in order to provide more stable employment conditions and improve job security for health workers.”
- “*Country* established Accountable Care Organizations (ACOs) to integrate care across different providers, aiming to improve patient outcomes and system efficiency.”

3.2 Please identify any recent reforms on system-wide health workforce staffing levels in your country/region:



Source(s), if available:



Box 9: Examples for planned reforms

- “The department introduced new rules mandating community services to have a registered nurse on duty 24 hours a day by 2026”.
- “The NHS is developing a 10-year health plan to address workforce shortages and improve healthcare delivery. This plan includes increasing training places for doctors.”
- “The xxx government has announced plans to revamp the NHS by increasing medical appointments.”

3.3 Please identify any future reforms on system-wide health workforce staffing levels in your country/region:



Please mention if your country/region is planning on defining staffing levels at the sectoral level in the future:



Source(s), if available:



Thank you for completing Part 1 of the survey.

Please complete the table below, before completing Part 2 (SECTORAL) of the survey.

Please identify for which specific healthcare sectors your country will fill in separate SECTORAL questionnaires:

Sectoral questionnaire	I will fill in the questionnaire	A colleague will fill in the questionnaire	We will not fill in the questionnaire
Hospitals	☐	☐ Name:	☐ Reason:
Primary care	☐	☐ Name:	☐ Reason:
Ambulatory care	☐	☐ Name:	☐ Reason:
Other sector	☐	☐ Name:	☐ Reason:

If you choose NOT to complete a subsequent SECTOR questionnaire because your country/region does not have defined staffing levels for this sector, please describe the reasons for NOT defining staffing levels:



Source(s), if available:



Staffing levels in European health systems: Why are they used, how are they defined, where are they implemented?

2

A survey co-designed by the European Observatory
on health systems and policies and the HSPA Expert Group
– working group on healthcare workforce

A focus on specific HEALTHCARE SECTORS



TABLE OF CONTENTS

Introduction	3
Section one: Approaches to and rationale for staffing levels	5
Section two: Governance	13
Section three: Evaluations, recent and planned reforms	22

INTRODUCTION

Background to the questionnaire

High-performing health systems rely on having the right number and mix of health workers in place to achieve health system goals and objectives, such as those related to health outcomes, staff and patient satisfaction, access, and quality. Shortages or imbalances in staffing can lead to delays, unsafe treatments, and poor user experiences, especially in ambulatory care clinics, emergency departments, and primary care settings to name a few. In the frame of Health System Performance Assessments (HSPA), staffing levels can be used as a set of representative measures to link workforce availability to broader health system objectives (i.e. access to healthcare, quality and safety of care, health outcomes and population health, to name a few).

As countries embark on a Health System Performance Assessments (HSPA) it would be beneficial to identify one or several indicators that help systems understand the link between workforce availability and broader health system objectives, and highlight areas that require further investigation.

Currently, many European countries have or are seeking to define and implement staffing standards across various sectors of their health systems. However, the terminologies and approaches used to establish these standards may differ between (and within) countries and there is little systematic knowledge about what countries are doing or about the emerging trends in staffing-related reforms.

The HSPA Expert Group identified health workforce as a priority topic for 2024-2025. After discussions and presentations on the EU actions addressing health workforce, the Expert Group decided at the October 2024 plenary to carry out a survey on existing practices regarding staffing levels, and their impact on the policies addressing health workforce shortages, to organise a workshop and issue a conference report. At the request of the European Commission, the European Observatory on Health Systems and Policies, an observer in the Expert Group was requested to support the work.

Survey aim

The aim of this survey is to therefore map approaches and methods to address staffing levels in the healthcare workforce in EU Member States and in Norway. By understanding how staffing levels are defined, implemented, utilized and evaluated across countries, there is an opportunity to facilitate mutual learning and identify best practice cases. The survey has been developed following several rounds of consultation with the HSPA co-chairs and members of the HSPA working group on the healthcare workforce. It was emphasized that the survey should map different healthcare sectors (e.g. hospital, primary care, etc.) and distinguish between doctors and nurses, as well as consider other professions.

Survey instructions

This survey consists of **two parts**:

- **Part 1 (HEALTH SYSTEM)** is focused on defined **staffing levels at the health-system level** in EU Member States and in Norway. Part 1 should be completed before Part 2.
- **Part 2 (SECTORAL)** is focused on defined staffing levels in **specific healthcare sectors** such as hospitals, primary care, ambulatory care etc. Part 2 should be completed if staffing levels at the sectoral level differ from those at the health system level.

This questionnaire is on **PART 2 (SECTORAL)**.

Please complete this questionnaire for **one** healthcare sector at a time. If you would like to provide responses for multiple sectors, **please complete a separate questionnaire for each sector** by duplicating this form and answering all questions again.

Before answering questions in the three main sections, **please fill-in the two boxes below** to indicate for which country/region and which healthcare sector (e.g. hospitals, primary care, ambulatory care) you are answering this questionnaire.

Please indicate for which country or region you are responding:



Please specify for which healthcare sector are you responding:



We truly appreciate your time and thank you for sharing your expertise and experience. The analysis of the results will be part of a HSPA Expert Group report.

■ Section 1:

Approaches to and rationale for staffing levels

1.1 What focus and approach to staffing levels in [sector] is used in your country/region: fixed ratios, flexible staffing models, or none?

Instructions

Please indicate whether your country/region uses fixed staffing ratios, flexible staffing models or no defined staffing levels for [sector] (see Box 1 for definitions) by checking the box next to the most appropriate answer(s) ☐. Please also check the appropriate profession(s) this approach applies to. If several answers are relevant, please provide some explanation in section marked “further comments” by first clicking in the text box and then typing. Countries that adopt a combination of staffing ratios can also use the section marked “further comments” to elaborate on their answer. If there are sources available please include these in the box below labeled “source(s)”.

Box 1: Definitions

Countries adopt different approaches to defining staffing levels:

Professions: What professions are you referring to: Doctors, nurses, carers, etc.

Fixed ratio staffing levels usually predefined ratios, typically specifying the number of health workers or professionals relative to patients, population groups, hospital beds, geographical areas... These ratios may be legally mandated or outlined in recommendations or guidelines. Fixed ratios are often used in hospital care and may involve nuanced differentiations based on specific care needs or settings.

Flexible staffing models are defined based on a combination of supply and demand factors aligning health worker availability with the healthcare needs of the population. This approach offers flexibility to adjust to workforce dynamics and evolving needs.

Other countries have no defined staffing levels. In some countries staffing levels are not formally defined, and neither fixed ratios nor flexible staffing models are utilized.

Please select all answers that apply

☐ My country/region uses fixed staffing ratios in this sector for:	☐ My country/region uses flexible staffing models in this sector for:	☐ My country/region does not use defined staffing levels in this sector
☐ Doctors	☐ Doctors	
☐ Nurses	☐ Nurses	
☐ Other professions - please specify here: 	☐ Other professions - please specify here: 	

Further comments:



Source(s), if available:



➤ If your country/region does **not** define staffing levels at [sector] level, proceed to question 1.4 and provide reasons for why staffing levels have **not** been defined.

➤ If you are answering for hospitals, please also complete the table below. For **other sectors**, please go to question 1.2.

Please select all answers that apply

<i>My country/region uses fixed staffing ratios in this sector for:</i>	<i>My country/region uses flexible staffing models in this sector for:</i>	<i>My country/region does not use defined staffing levels in this sector</i>
☐ General care unit	☐ General care unit	☐ General care unit
☐ Intensive care units	☐ Intensive care units	☐ Intensive care units
☐ Emergency departments	☐ Emergency departments	☐ Emergency departments
☐ Surgical unit	☐ Surgical unit	☐ Surgical unit
☐ Maternity services	☐ Maternity services	☐ Maternity services
☐ Psychiatric unit	☐ Psychiatric unit	☐ Psychiatric unit
☐ Other medical specialty or ward. Please specify here: 	☐ Other medical specialty or ward. Please specify here: 	☐ Other medical specialty or ward. Please specify here:
☐ Fixed ratios	☐ Fixed ratios	

Further comments (for example about ratios, professions, alternative solutions):



Source(s), if available:



1.2 What are the reasons for defining staffing levels in [sector] in your country/region?

Instructions

In Box 2 below several goals are defined. These can be the reasons for defining staffing levels in a country/region. We have identified three broad categories:

- Health system objectives and goals
- Health workers related goals
- Societal goals.

To answer the following questions, please refer to official documents such as policies, legislation, and guidelines. These may be issued by the government, the Ministry of Health, competent arm's length bodies, social partners delegated to draft such documents. If, for example, **effectiveness of care** is an official reason as to why your country/region is defining staffing levels, please check the box next to it ☐.

Several official goals can be reported. If the documents you have identified include additional reasons for defining staffing levels, please specify these in your response. Please include in writing or provide a weblink or an attachment for the sources you reference.

Please select all answers that apply

Health system objectives and goals	Goals related to health workers	Societal goals – well-being
☐ Effectiveness of care	☐ Protecting the health and well-being of health workers	☐ Social cohesion
☐ Safety of care	☐ Increasing retention by improving working conditions	☐ Economic development
☐ User experience	☐ Don't know	☐ Environmental sustainability
☐ Access to care		☐ Don't know
☐ Patient centredness		
☐ Health improvement		
☐ Financial protection		
☐ Quality of care		
☐ Don't know		

Other relevant health system objectives and goals:



Continued on next page >>

Other relevant health workers related goals:



Other societal goals:



Source(s), if available:



Box 2: Definition of declared objectives and goals for defining staffing levels

Health system objectives and goals: they include people-centeredness, health improvement, and financial protection. These goals are the outcomes used to measure the performance of a health system. They are very relevant to patients and citizens. The health workforce influences these final goals by impacting the intermediate objectives.

Effectiveness of care: The ability of an intervention to have a meaningful effect on patients in normal clinical conditions.

Safety of care: The prevention of errors and adverse effects associated with health services.

User experience: Users' beliefs, preferences, perceptions, responses, and behaviors that occur before, during and after.

Box 2 continued on next page >>

Access to care: The opportunity or ease with which individuals or communities are able to use appropriate health services.

Patient centeredness: As a goal which captures how far the health system adequately addresses people's non-medical health needs.

Health improvement: A fundamental goal of any health system. It encompasses the health of the population at different points in the life cycle and is measured through morbidity and mortality indicators as well as aggregate life expectancy.

Financial protection: Refers here to the health system's role in protecting the population against risk. For example, if there are not enough doctors working in the public sector, people may be forced to visit a doctor in the private sector, leading to out-of-pocket spending.

Quality of care: is used as a generic term comprising various dimensions including e.g. access, safety, effectiveness, user experience but also efficiency and equity.

Goals related to health workers: These goals focus on safeguarding the well-being of health workers, ensuring their job satisfaction, and creating a supportive work environment that enables them to thrive in their roles.

Protecting the health and well-being of health workers: adequate staffing level are a key to ensure that health workers are not exposed to constant work overload affecting their health.

Improving retention by improving working conditions: adequate staffing levels are an instrument to retain health workers in healthcare organizations.

Societal Goals: These goals are outside the direct scope of the health system but are still closely linked to staffing levels. For example, the environmental impact of health workers' green skills can contribute to sustainability. A well-staffed health workforce can also positively affect economic development by supporting a healthier population, which in turn contributes to a stronger economy. Secure staffing levels during times of economic crisis have been shown to help maintain economic stability. Additionally, staffing levels can affect social cohesion, especially in areas with limited access to care, such as medical deserts or deprived urban areas.

Social cohesion: to achieve collective commitment to unity, mutual support, and shared responsibility among individuals and communities. This goal encompasses eliminating discrimination, reducing income inequality, and ensuring equal access to healthcare.

Economic development: is a vital societal goal which enables higher living standards, job creation, and increased innovation.

Environmental sustainability: refers to the interconnectedness of environmental concerns with broader sustainability objectives. It refers to a balanced and inclusive approach to economic growth that seeks to meet the needs of present and future generations without compromising natural resources and global ecosystems.

NOTE Intermediate Objectives:

These objectives are directly related to how well health systems deliver services and achieve overall health outcomes. They include effectiveness, safety, user experience, and access to care. Achieving these objectives is crucial for high-quality service delivery. For this to happen, health systems need the right number of skilled workers in the right places at the right times. These intermediate objectives are a key link between staffing levels and the final health system goals. If intermediate objectives aren't met, it's unlikely the system will perform well in terms of people-centered care, health improvement, and financial protection. The relationship between staffing levels and health system outcomes can be understood through these intermediate objectives.

1.3 What were the reasons for choosing one approach (e.g., fixed ratios or flexible staffing models) over the other?

Please provide the reasons for choosing one approach (e.g., fixed ratios or flexible staffing models) over the other. To guide your response, here are a couple of examples (see Box 3):

Box 3: Reasons for preferences – examples

- “We have chosen fixed ratios because it is much easier to define and it provides enough orientation for recruitment and health workforce development”.
- “Flexible staffing models are too complicated and necessitate too much research.”
- “Flexible staffing models take local variations into account and define much better the health workforce needed at local/provider level”.
- “A consensus among experts suggested it was the best approach”.
- “The choice was informed by scientific evidence”.

Why has your country/region chosen one approach to define staffing levels over the other?



Source(s), if available:



➞ Please now **go to section 2** – governance.

1.4 Why are staffing levels not defined?

Instructions

This is an open-ended question. Please provide reasons why staffing levels have not been defined in your country/region for [sector]. To guide your response, here are a couple of examples (see Box 4):

Box 4: Examples for reasons not to define staffing levels

- “The government felt there is not enough evidence for supporting the definition of staffing levels”.
- “The government was worried that ‘one-size-fits-all’ would harm already well-established individual staffing levels at local or provider level”.
- “If staffing levels are defined, those providers which exceed the levels may decrease their workforce”.

If it is difficult to base your response on an official document, please use any relevant knowledge you can obtain and specify the sources you reference. If no sources are available input from experts or stakeholder views are also welcome.

Reasons for NOT defining staffing levels



Source(s), if available:



➞ If staffing levels are **not defined** for [sector] in your country/region, **please go to section 3.**

■ Section 2: Governance

In this section we would like to understand the process of defining and implementing staffing levels.

2.1 Who sets the staffing levels in [sector] in your country/region?

Staffing levels can be set by the government or the Department/Ministry of Health. In some countries, this responsibility may also be delegated to a specialized authority. We would like to understand the roles and interactions of the various stakeholders involved in setting staffing levels for [sector] in your country/region. These may include professional associations, local authorities, and in some cases expert advisory councils or social partners. Please provide details on how these entities collaborate or influence the decision-making process related to staffing levels in [sector].

Who sets the staffing levels in [sector] in your country/region?



Source(s), if available:



2.2 What tools and methods are used to determine staffing levels in [sector] in your country/region?

When staffing levels are defined, scientific tools and methods are often used to determine supply and demand (see Box 5 for definitions). This can apply to both fixed ratios and flexible staffing models. **Please check the boxes on the next page** that apply, or fill in what you mean by “other”.

Box 5: Description of terminology

Population profiling: the analysis of demographic, epidemiological, and socio-economic characteristics of a population to inform health workforce planning.

Health service usage/workload profiling: analyzes how healthcare services are utilized and the workload distribution among health workers. Helps define optimal staffing levels by ensuring that workforce supply matches service demand effectively.

Provider/local health workforce profiling: the assessment of the composition, distribution, and capacity of health workers within a specific geographic area or healthcare system. Ensures that staffing levels are aligned with local population health needs, service demand, and healthcare delivery capacity.

Clinical guidelines: recommendations that outline best practices for defining staffing levels by determining the type, number, and mix of health workers required to deliver high-quality care efficiently.

Modelling approaches: refer to the use of mathematical, statistical, and computational techniques to estimate and predict staffing levels based on healthcare demand, workforce supply, and system dynamics. Help policymakers and planners make informed decisions about workforce distribution, recruitment, and training needs.

Protocols: standardized procedures or guidelines that help determine staffing levels by specifying the required number, skill mix, and distribution of professionals needed to deliver care efficiently and safely.

Please select all answers that apply

- | | |
|---|--|
| ☐ None | ☐ Modelling approaches |
| ☐ Population profiling | ☐ Protocols |
| ☐ Health service usage/workload profiling (acuity based assessments) | ☐ Other, please specify |
| ☐ Provider/local health workforce profiling | ☐ Don't know |
| ☐ Clinical guidelines | |

Further comments:



Source(s), if available:



2.3 Are defined staffing levels for [sector] in your country/region legally binding?

In some countries/regions staffing levels may be mandatory without a legally binding instrument, especially in national health services, where there is a clear line of command from the central government to local providers. In contrast, systems with private or independent providers often rely on legally binding regulations for mandatory defined staffing levels, although enforcement may not always be directly handled by the Ministry of Health. In such systems, enforcement could come through the courts, initiated by health workers, citizens, or local authorities. In health systems with strong social partnerships, staffing levels may be governed by contractual agreements that establish legally binding obligations. Other countries may opt for a more flexible approach, relying on transparency and good faith rather than legal enforcement.

Please specify if staffing levels for [sector] in your country/region are legally binding:

- ☐ Yes
- ☐ No
- ☐ Don't know

Further comments (i.e. what is the legal status, what type of instrument are you using if staffing levels are not legally binding):



Source(s), if available:



2.4 How is compliance with staffing levels for this [sector] monitored?

We would like to understand how the implementation of staffing levels in [sector] is monitored and who is responsible for doing so. Monitoring adherence to fixed staffing ratios is typically straightforward, as it involves tracking whether the established ratios are met. However, monitoring flexible staffing levels is more complicated, because it requires verifying that the rules are being correctly applied.

What are the monitoring requirements for the implementation of the staffing levels?



Source(s), if available:



2.5 What sanctions and/or incentives are in place for meeting or not meeting staffing requirements for [sector]?

We would like to understand whether compliance is incentivized or penalized. While transparency can help ensure compliance, it may not suffice. Sanctions both positive and negative may play a role in encouraging adherence. Negative sanction may be imposed for non-compliance, while positive incentives may be used to encourage better performance. We are interested in how these mechanisms are implemented and their effectiveness.

What are the sanctions/incentives that apply for failing to achieve or for achieving the staffing levels?



Source(s), if available:



2.6 To what extent can providers at regional or local levels adjust national staffing levels?

This question seeks to understand the role of the regional and local/provider-level authorities in determining staffing levels. There are often significant regional and local differences in both population demographics and the labour markets for healthcare workers. These differences can exist between urban and rural areas but also between affluent and underserved regions. We would like to know the extent to which regional or local/provider-level entities have the flexibility to adjust national staffing levels?

To what extent can providers at regional or local levels adjust national staffing levels?



Source(s), if available:



2.7 Defined staffing levels and health workforce planning for [sector]: is there a connection, and if so, what does it look like?

Setting defined staffing levels can directly impact the number of health workers needed as well as the required skill-mix. Adjustments in staffing may require planning for an increase in the broader healthcare workforce or a focus on specific professions. We are interested in understanding whether there is a connection between staffing levels and health workforce planning in your country/region. For example, introducing minimum staffing requirements in specific wards may require targeted incentives for relocation and retention, or a reduction in non-clinical staff. If such a connection exists, what mechanisms or processes are in place to align staffing levels with health workforce planning needs?

2.7a Does a connection between defined staffing levels in [sector] and health workforce planning exist in your country/region?

- ☐ Yes
- ☐ No
- ☐ Don't know

➔ If you answered "no" or "don't know", please go to question 2.8.

2.7b If such a connection exists, what mechanisms or processes are in place to align staffing levels with health workforce planning needs?



Source(s), if available:



2.8 Staffing levels and investment in the health workforce for [sector]: is there a link, and if so, what does it look like?

Changes in staffing levels can also have significant implications for investment in the health workforce, as they may lead to a change in the number of workers or a shift in the required skill mix. This, in turn, could require additional financial resources for salaries, training, and other workforce-related investments. We would like to understand whether there is an established connection between staffing levels and investment in the health workforce. If such a connection exists, how is it reflected in budgeting or resource allocation for salaries, training, and other workforce needs?

2.8a Does a connection between defined staffing levels in [sector] and health workforce investment exist in your country/region?

- ☐ Yes
- ☐ No
- ☐ Don't know

➡ If you answered "no" or "don't know", please go to question 2.9.

2.8b If such a connection exists, how is it reflected in budgeting or resource allocation for salaries, training, and other workforce needs?



Source(s), if available:



2.9 Staffing levels and training capacities for [sector]: is there a link, and if so, what does it look like?

Changes in staffing levels may impact the need for training capacities at universities and vocational training institutions. As the demand for healthcare workers fluctuates, training programs must be adjusted to ensure alignment with these needs. This could involve close collaboration with various ministries, such as those in charge of science, education, as well as with universities, medical schools, and other professional training institutions. We are interested in understanding whether there is a connection between staffing levels and training capacities, and how this connection is managed across different sectors and institutions.

2.9a Does a connection between defined staffing levels in [sector] and health workforce training exist in your country/region?

- ☐ Yes
- ☐ No
- ☐ Don't know

➔ If you answered "no" or "don't know", please go to section 3.

2.9b What type of connection between defined staffing levels and health workforce training exists?



Source(s), if available:



Section 3:

Evaluations, recent and planned reforms

In this section, we ask you to identify existing evaluations, recent reforms and future plans or policies related to health workforce staffing levels in your country/region for [sector]. For **evaluations**, please consider highlighting any evaluation methods used by your country (see Box 6 for examples) For **recent reforms**, please consider highlighting any significant changes, updates, or new policies implemented in the last 5–10 years that impact staffing levels (see Box 7 for examples). For **future plans or policies**, please describe any planned initiatives, strategies, or policies under development aimed at addressing staffing levels (see Box 8 for examples).

Box 6: Examples for existing evaluations

- “The National Institute for xx has developed workforce planning tools that evaluates patient acuity and dependency to determine staffing levels.”
- “Institute xyz in country abc evaluates workload data to determine staffing requirements in hospitals and primary care settings.”

3.1 Please identify any existing evaluations of health workforce staffing levels for [sector] in your country/region:



Source(s), if available:



Box 7: Examples for recent reforms

- “My country implemented a legislation introducing fixed staffing ratios to determine staffing levels in health systems in 20...”
- “In 2020, my country, has reformed the Framework Statute for Health Personnel, setting limits on temporary work in order to provide more stable employment conditions and improve job security for health workers.”
- “Country established Accountable Care Organizations (ACOs) to integrate care across different providers, aiming to improve patient outcomes and system efficiency.”
- “In 2020, country, introduced a model focusing on family medicine to enhance primary care delivery and patient-centered services.”

3.2 Please identify any recent reforms on health workforce staffing levels for [sector] in your country/region:



Source(s), if available:



Box 8: Examples for planned reforms

- “The department introduced new rules mandating community services to have a registered nurse on duty 24 hours a day by 2026”.
- “The NHS is developing a 10-year health plan to address workforce shortages and improve healthcare delivery. This plan includes increasing training places and focusing on out-of-hospital care and illness prevention to reduce the burden on hospitals.”
- “Starting in 2023, country plans to prioritize hospital reforms focusing on improving emergency care, addressing nursing shortages, and restructuring hospital financing and planning.”
- “The xxx government has announced plans to revamp the NHS by increasing medical appointments and providing additional funding for general practices.”

3.3 Please identify any future reforms on health workforce staffing levels for [sector] in your country/region:



Source(s), if available:



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